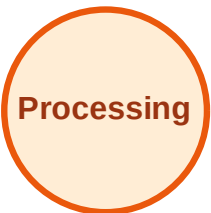


BFS Traversal

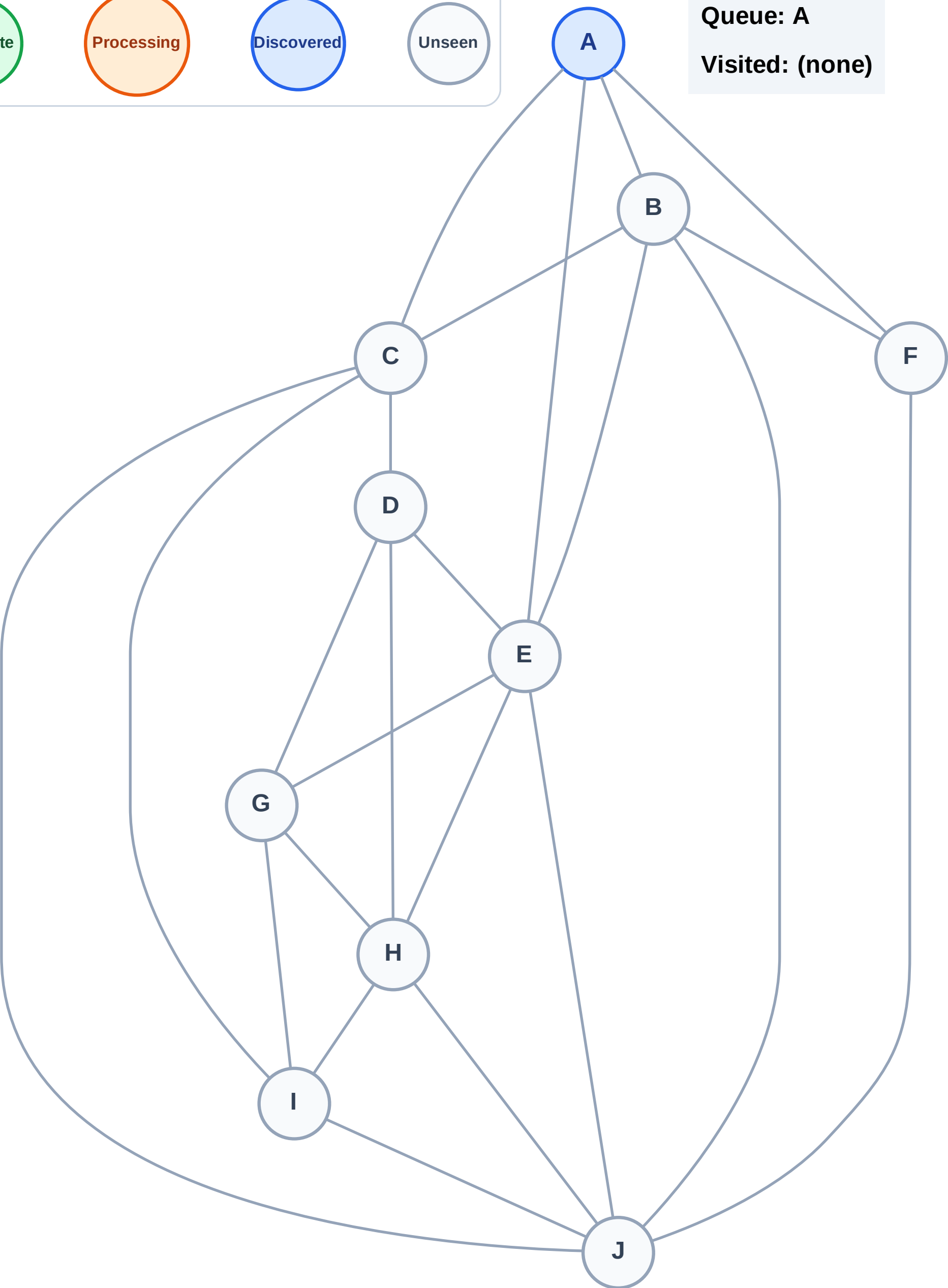
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

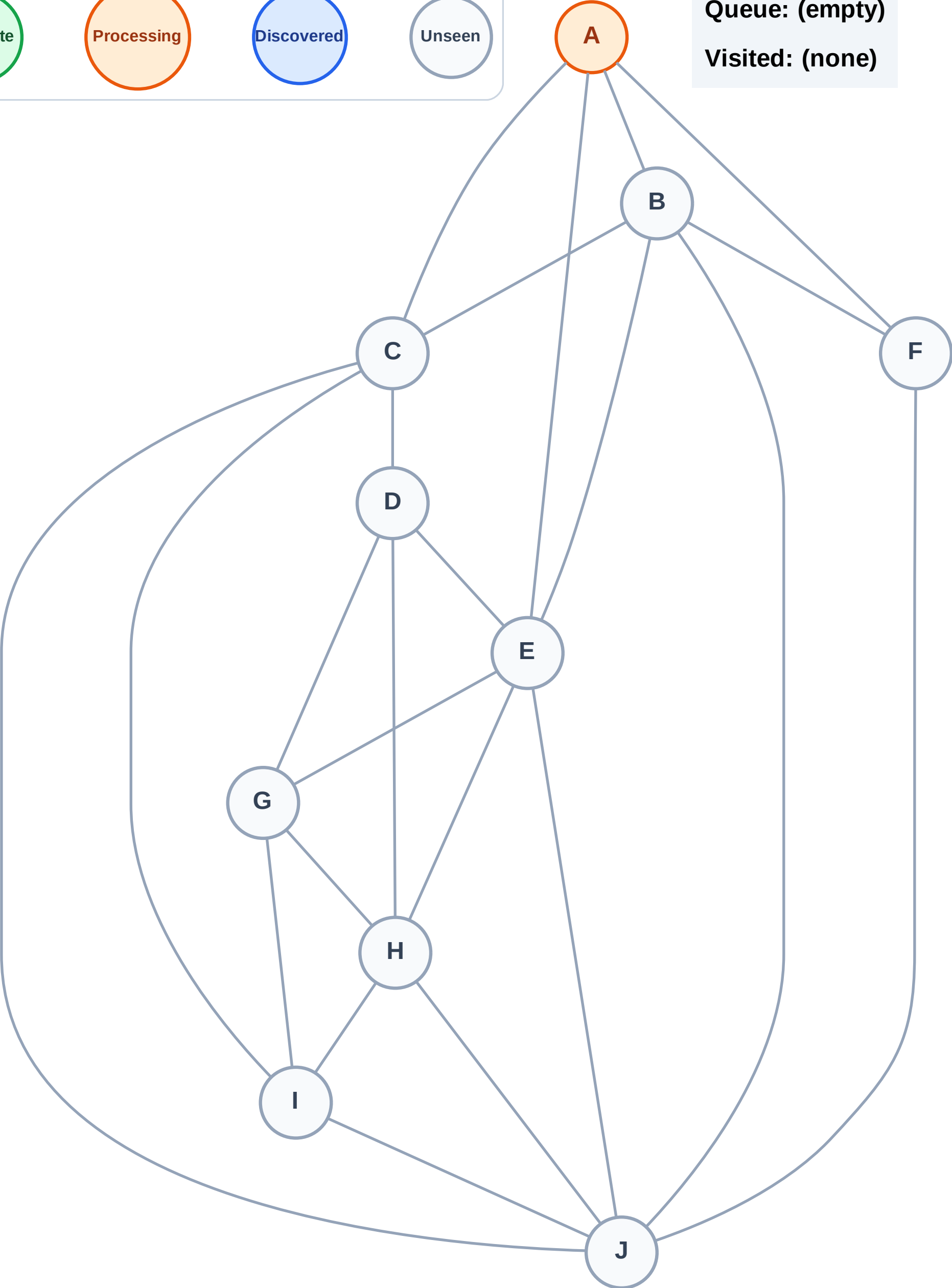
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, C, E, F from A

Legend

Complete

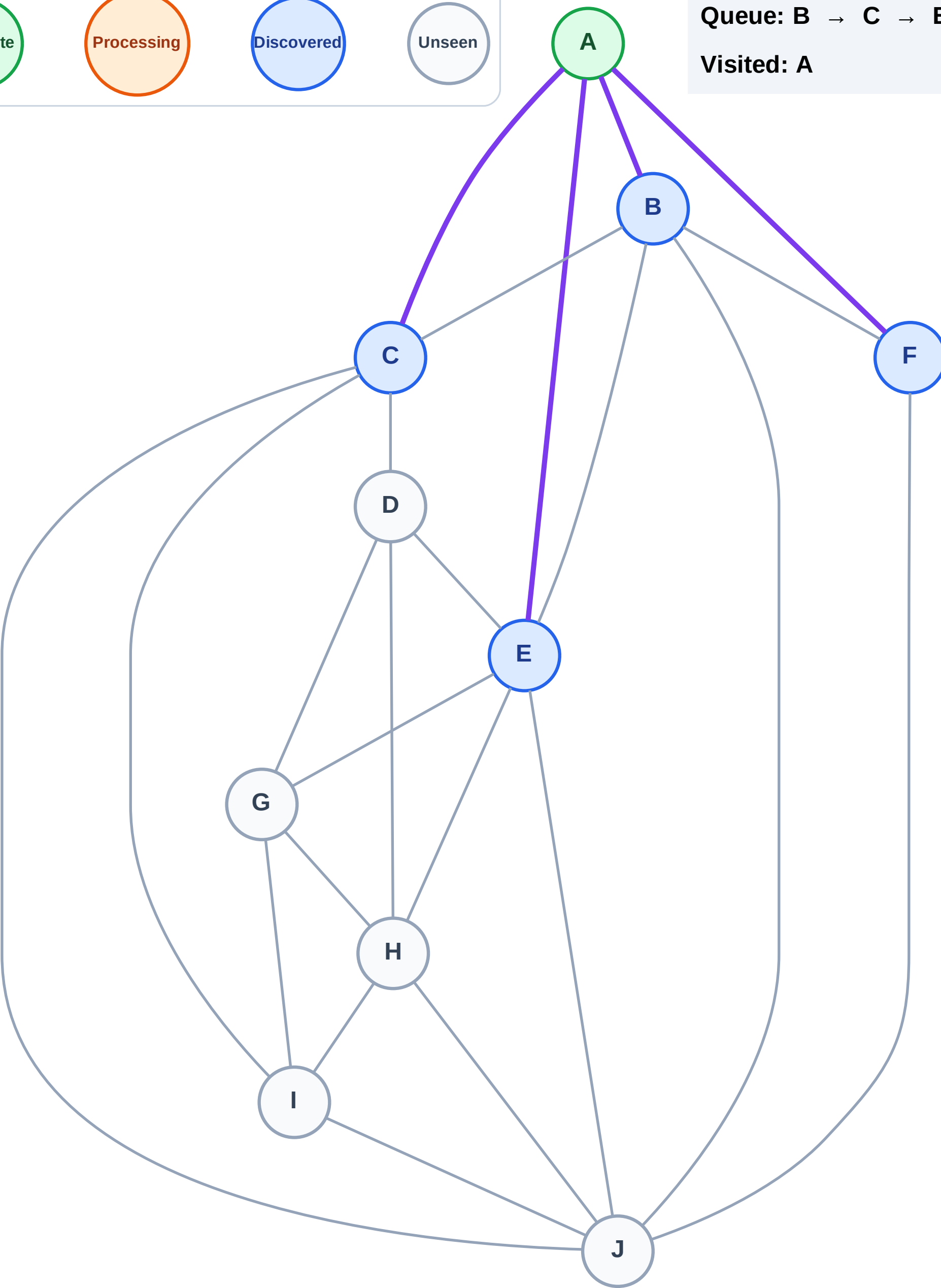
Processing

Discovered

Unseen

Queue: B → C → E → F

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

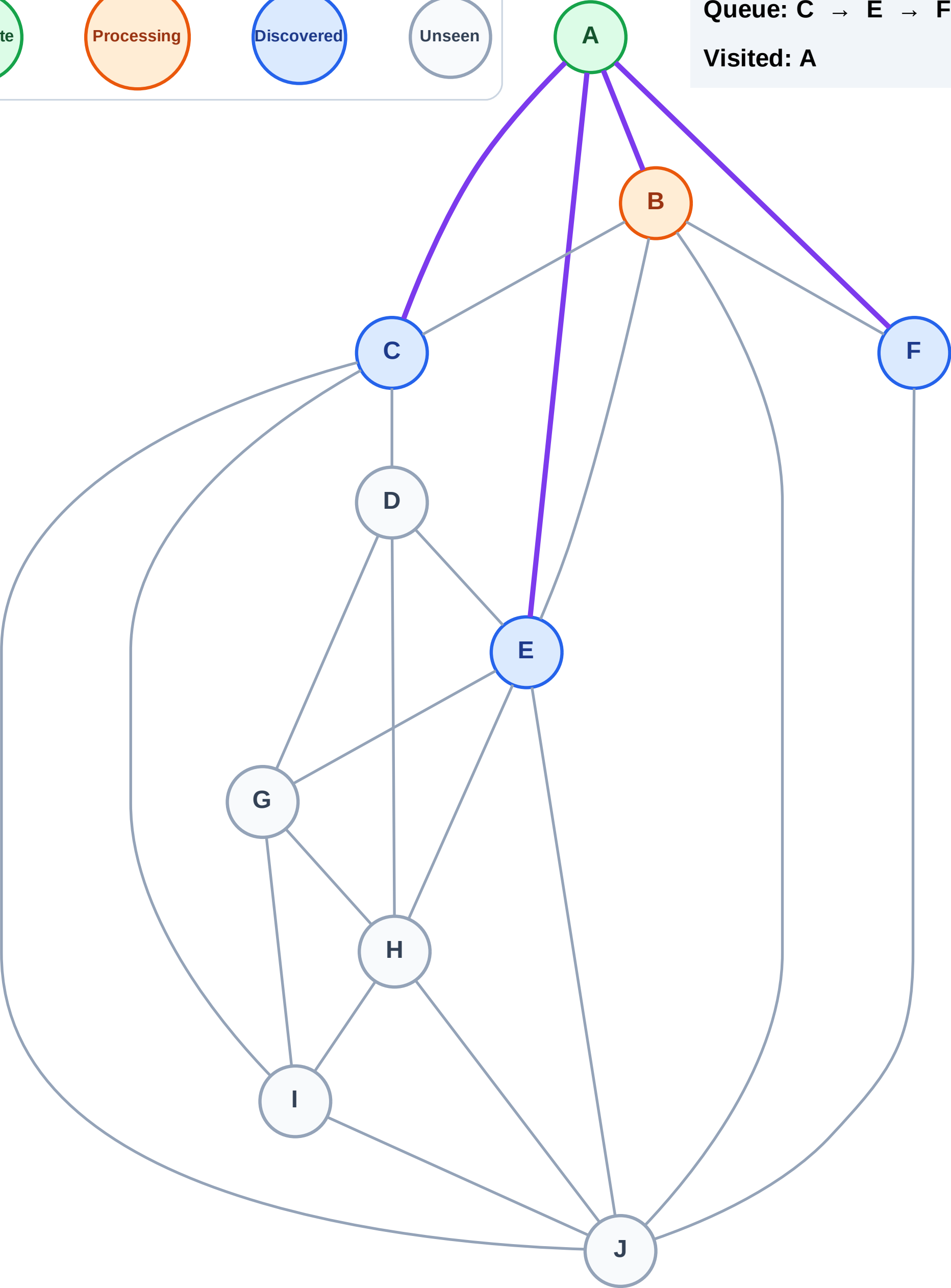
Processing

Discovered

Unseen

Queue: C → E → F

Visited: A



BFS Traversal

Step 5: Discover J from B

Legend

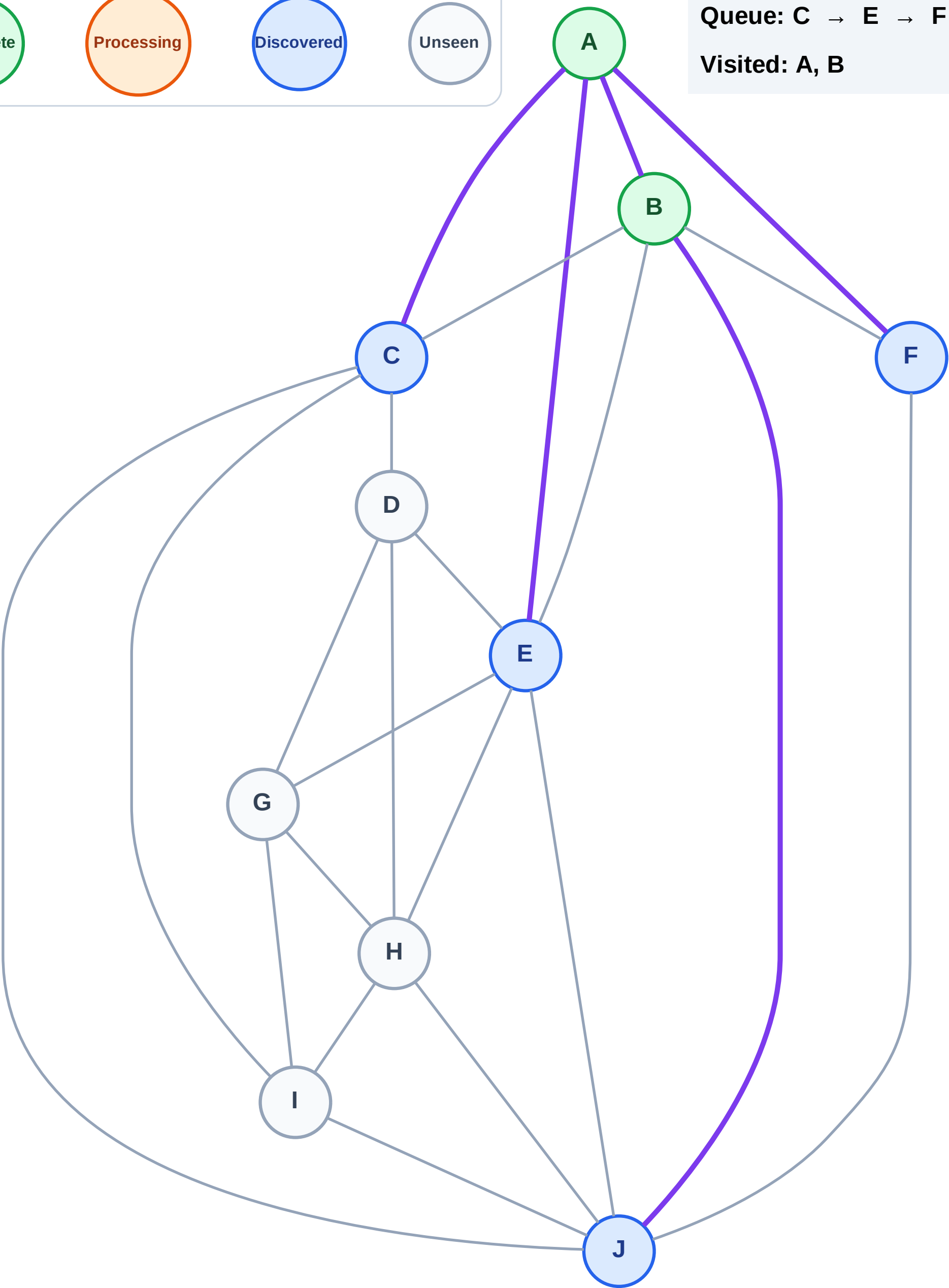
Complete

Processing

Discovered

Unseen

Queue: C → E → F → J
Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

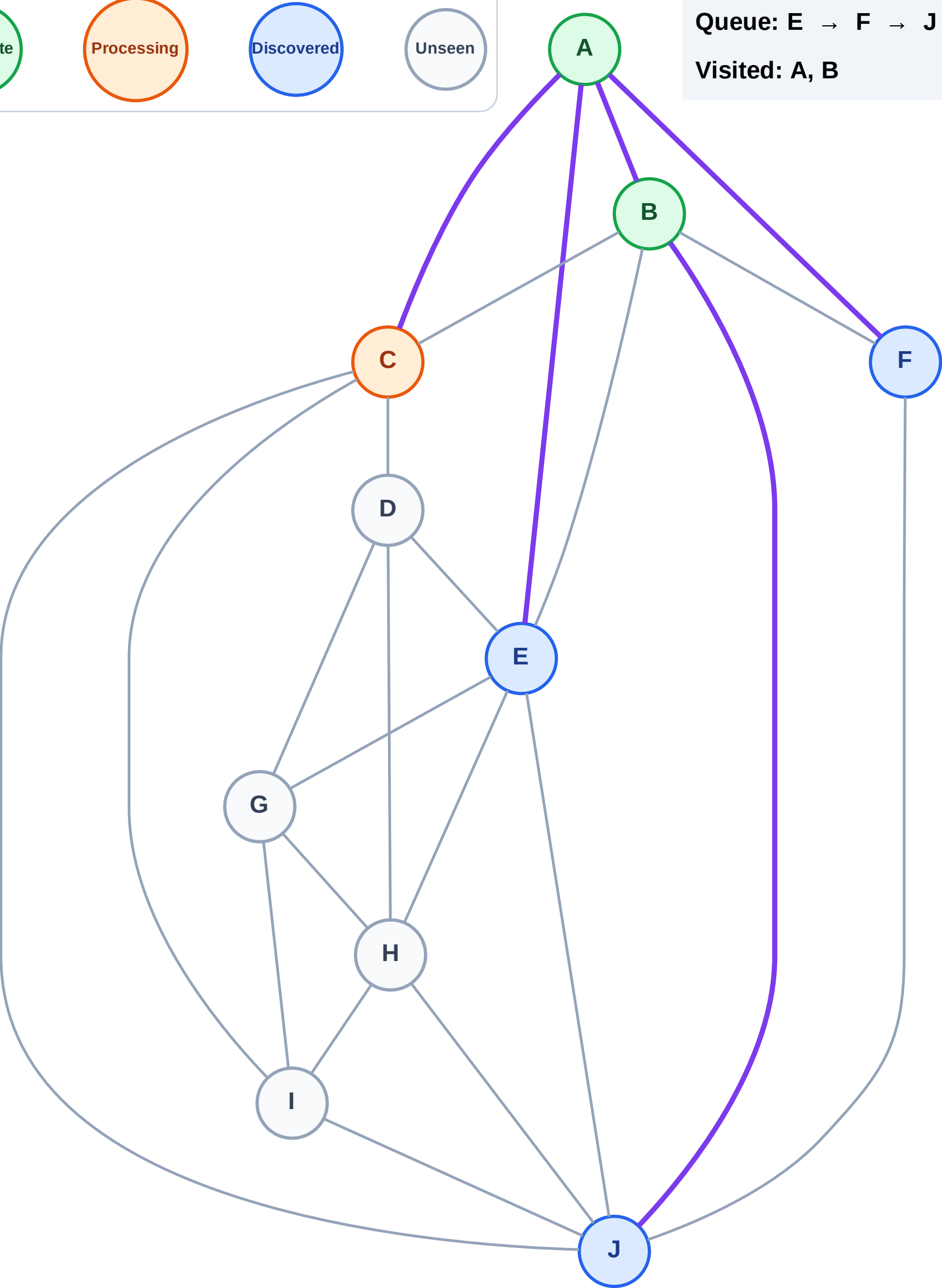
Processing

Discovered

Unseen

Queue: E → F → J

Visited: A, B



BFS Traversal

Step 7: Discover D, I from C

Legend

Complete

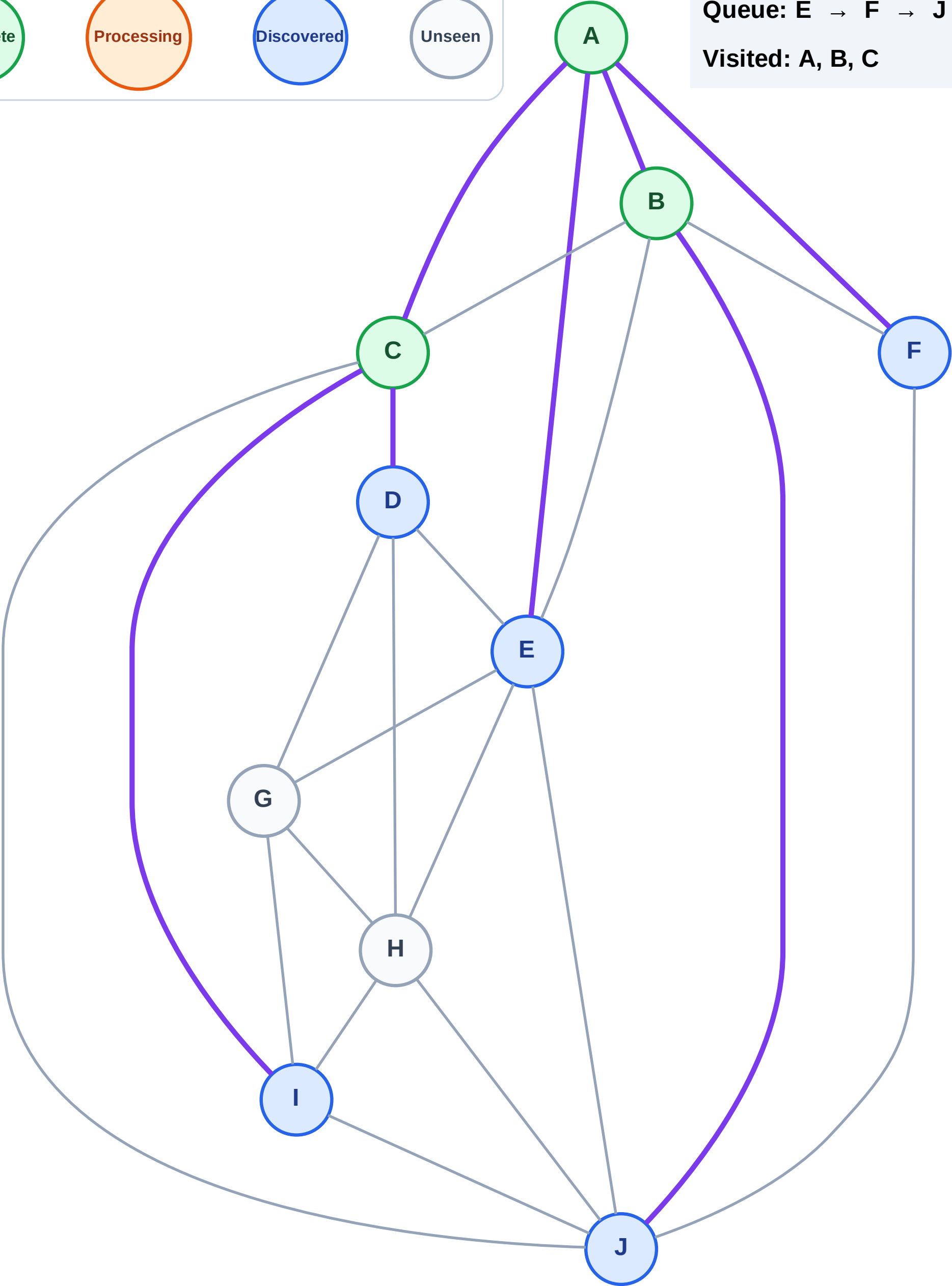
Processing

Discovered

Unseen

Queue: E → F → J → D → I

Visited: A, B, C



BFS Traversal

Step 8: Process node E

Legend

Complete

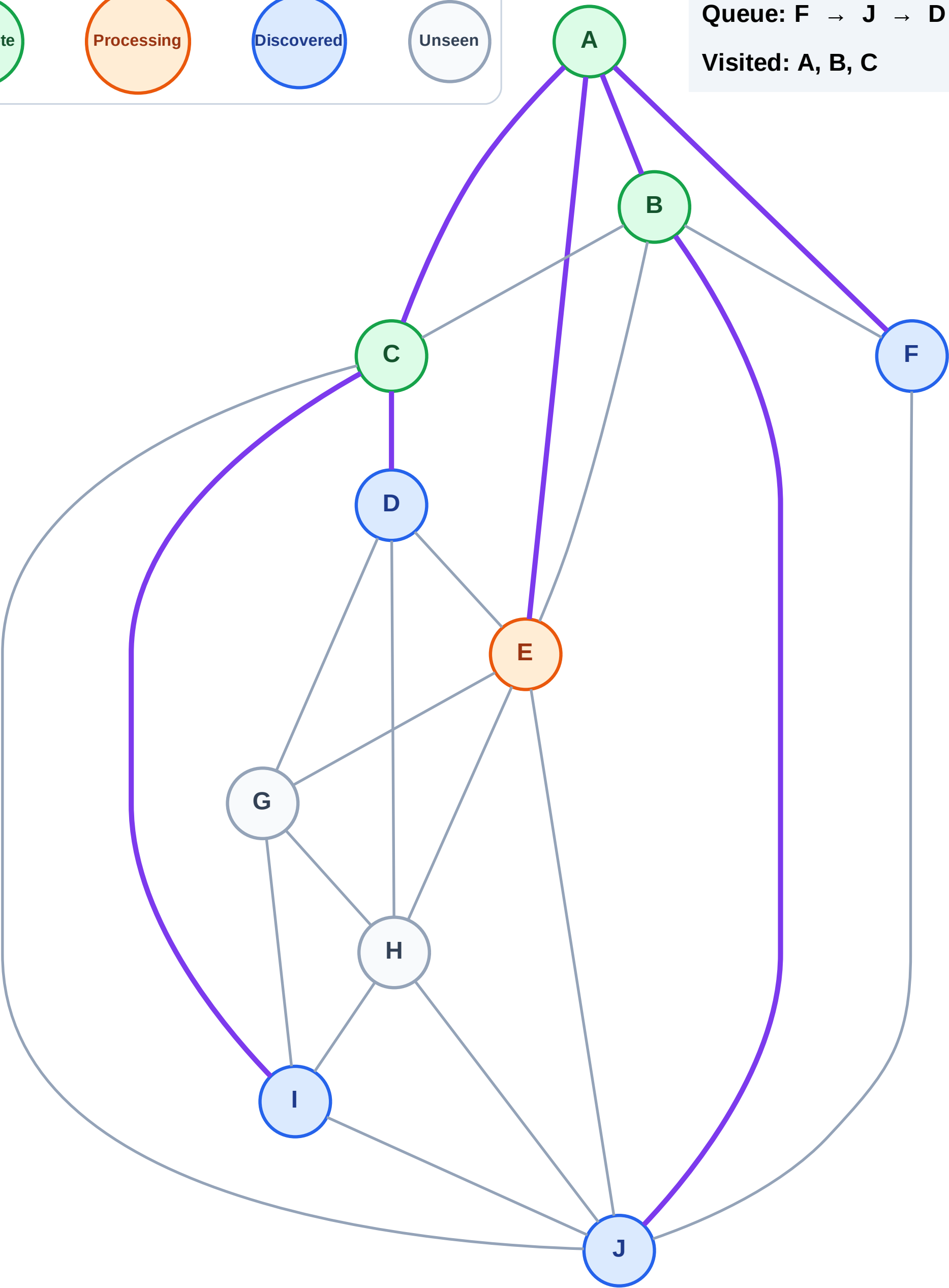
Processing

Discovered

Unseen

Queue: F → J → D → I

Visited: A, B, C



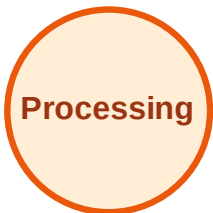
BFS Traversal

Step 9: Discover G, H from E

Legend



Complete



Processing



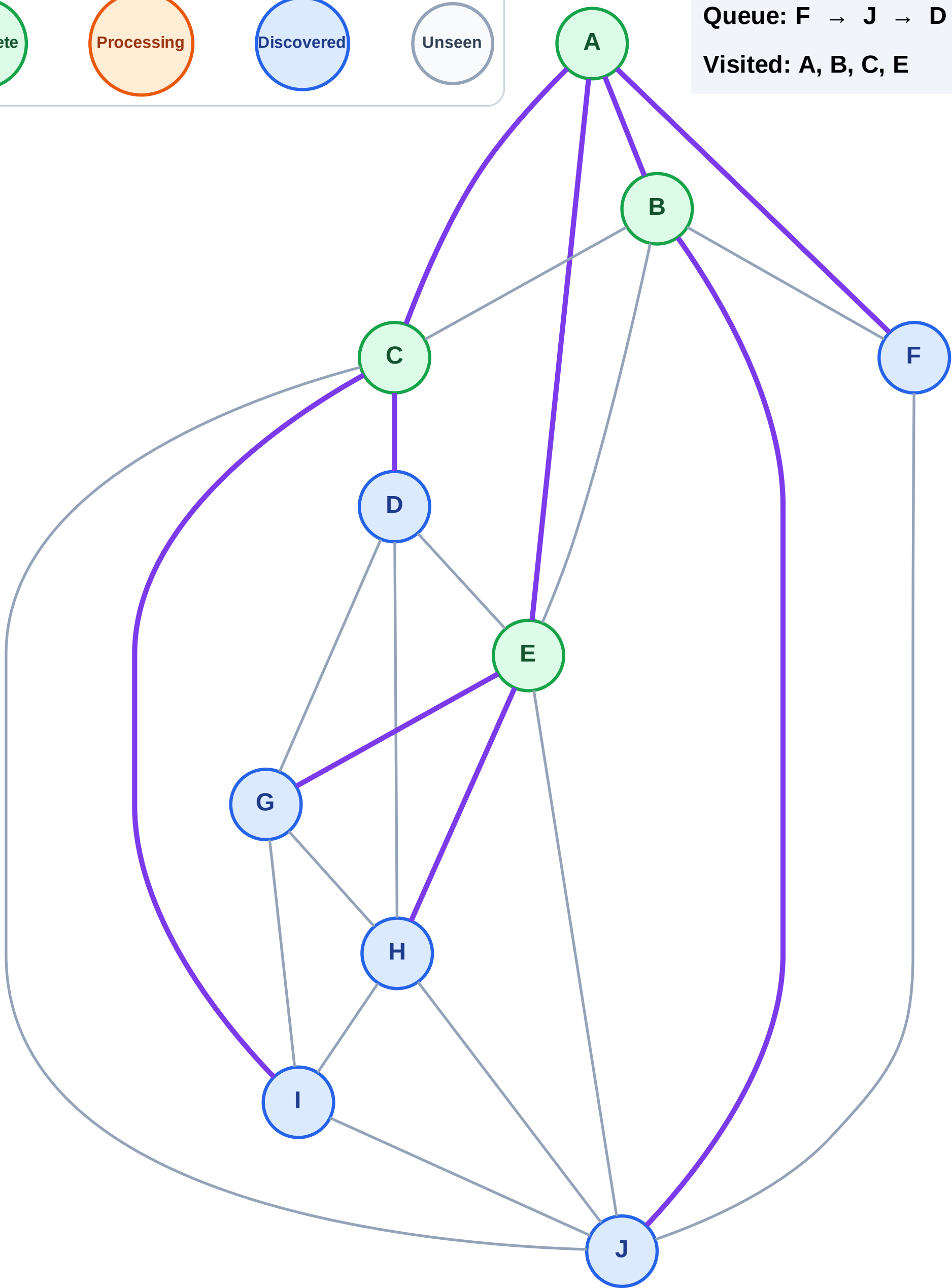
Discovered



Unseen

Queue: F → J → D → I → G → H

Visited: A, B, C, E

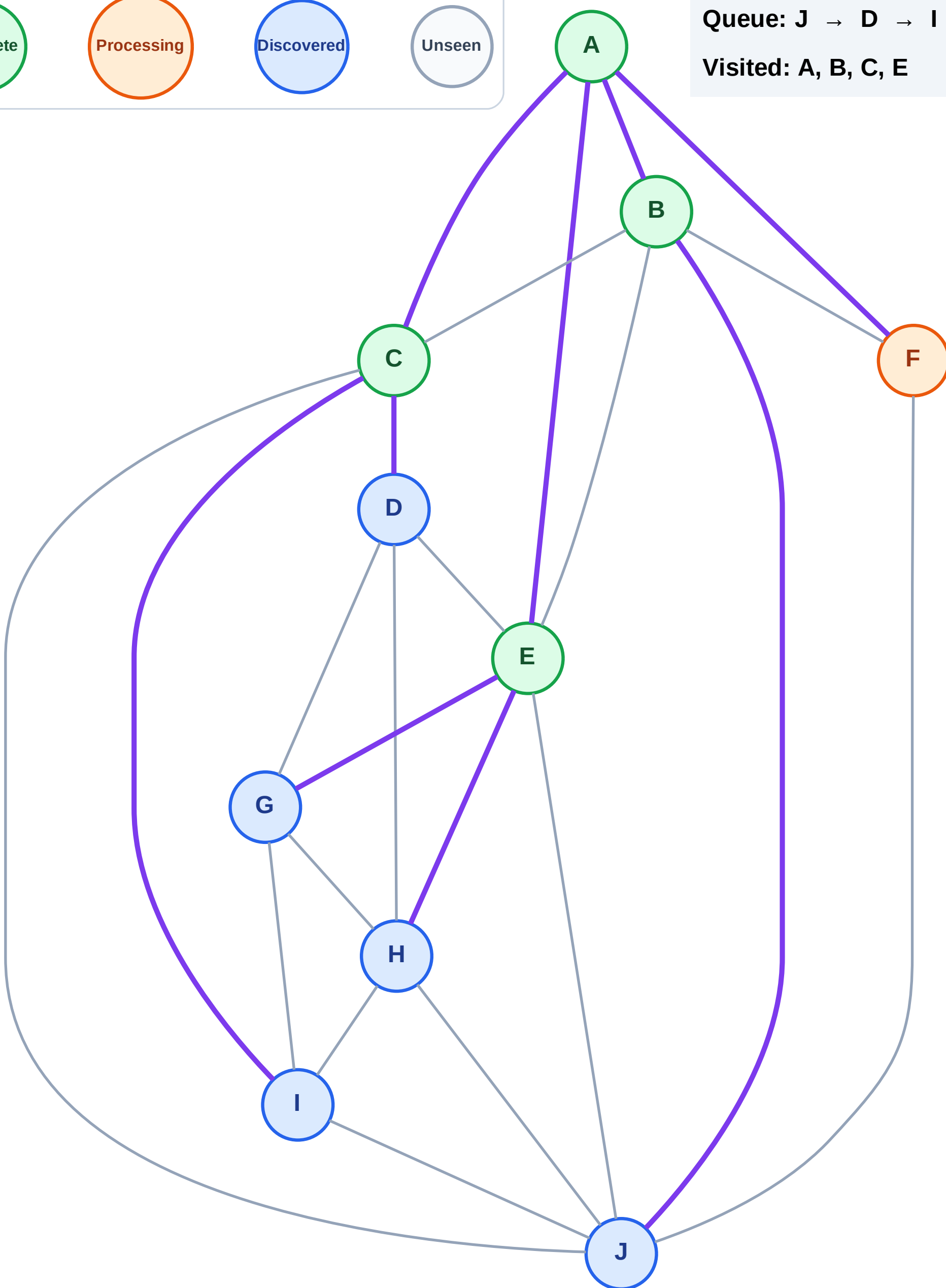


Step 10: Process node F

Legend

Unseen

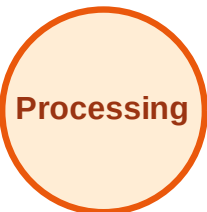
Visited: A, B, C, E



BFS Traversal

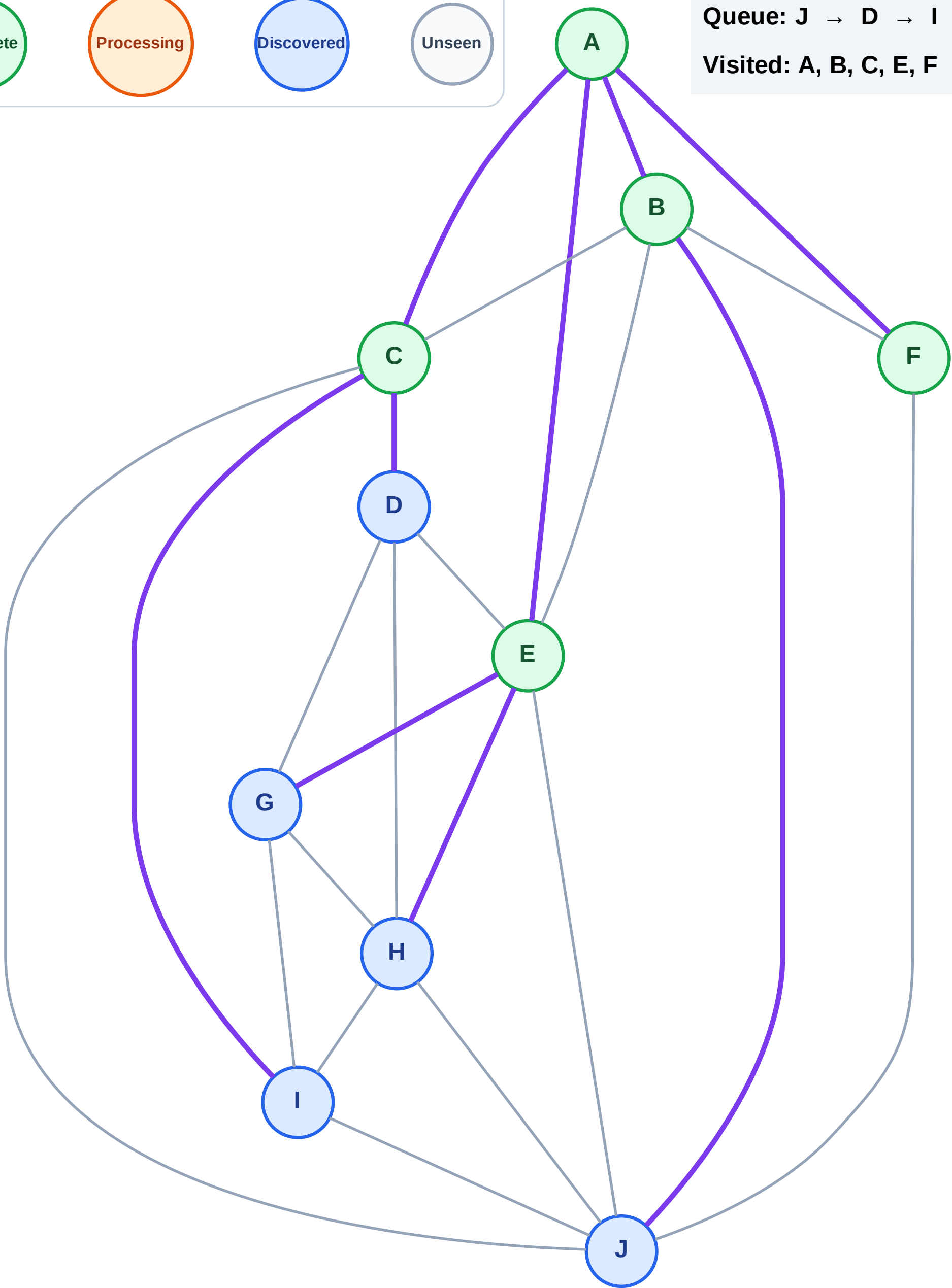
Step 11: No new neighbors from F

Legend



Queue: J → D → I → G → H

Visited: A, B, C, E, F



BFS Traversal

Step 12: Process node J

Legend

Complete

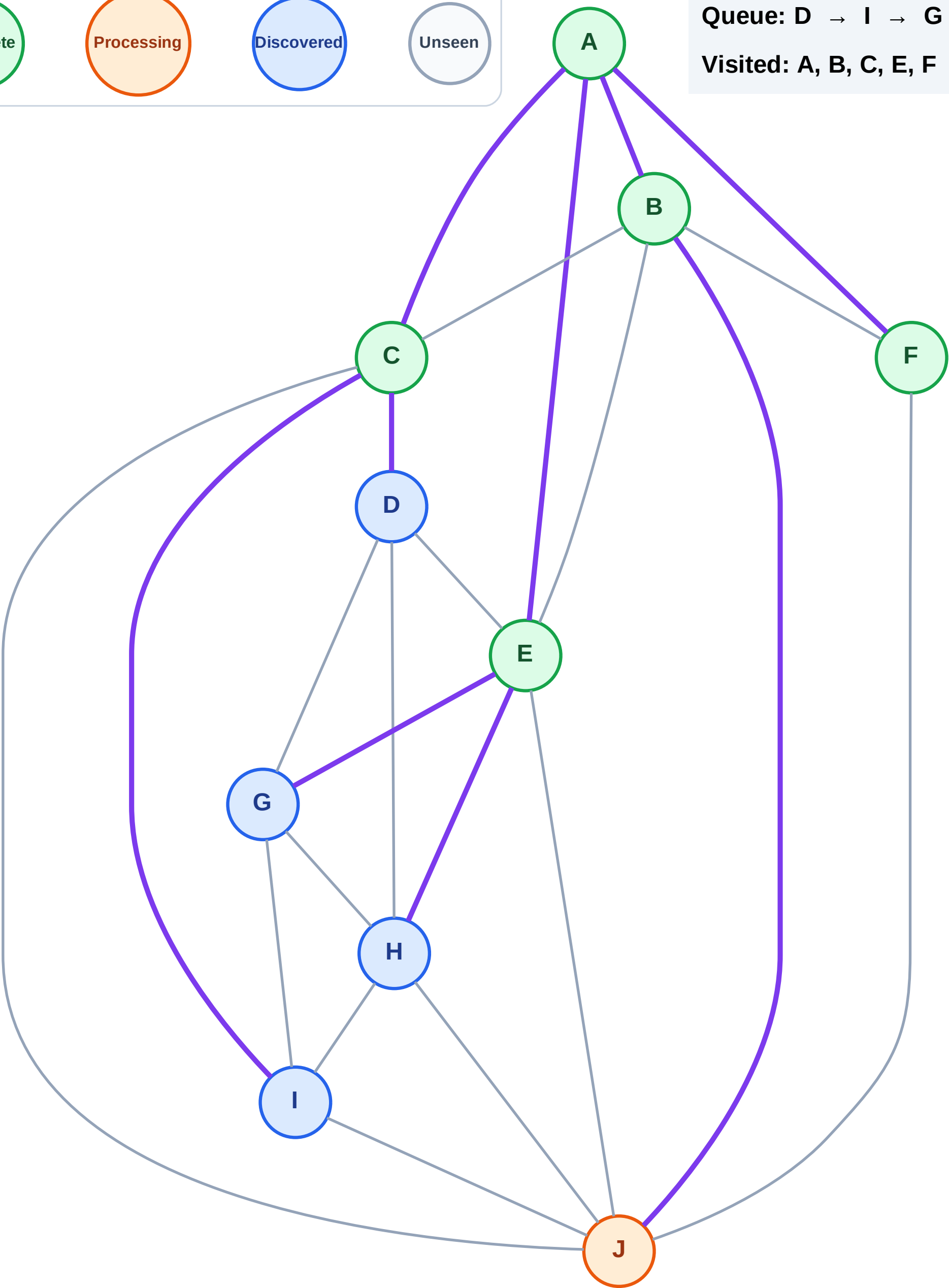
Processing

Discovered

Unseen

Queue: D → I → G → H

Visited: A, B, C, E, F



BFS Traversal

Step 13: No new neighbors from J

Legend

Complete

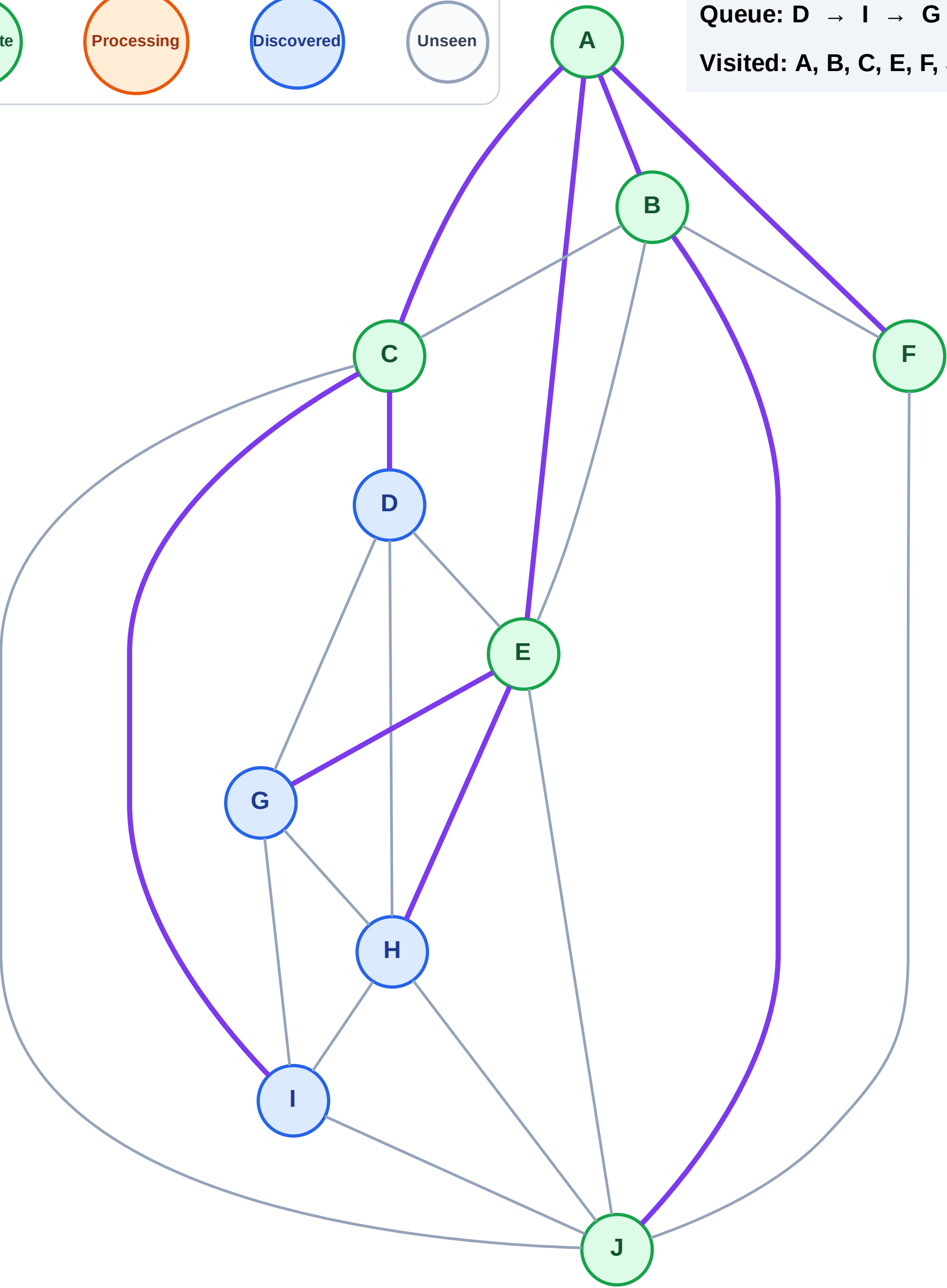
Processing

Discovered

Unseen

Queue: D → I → G → H

Visited: A, B, C, E, F, J

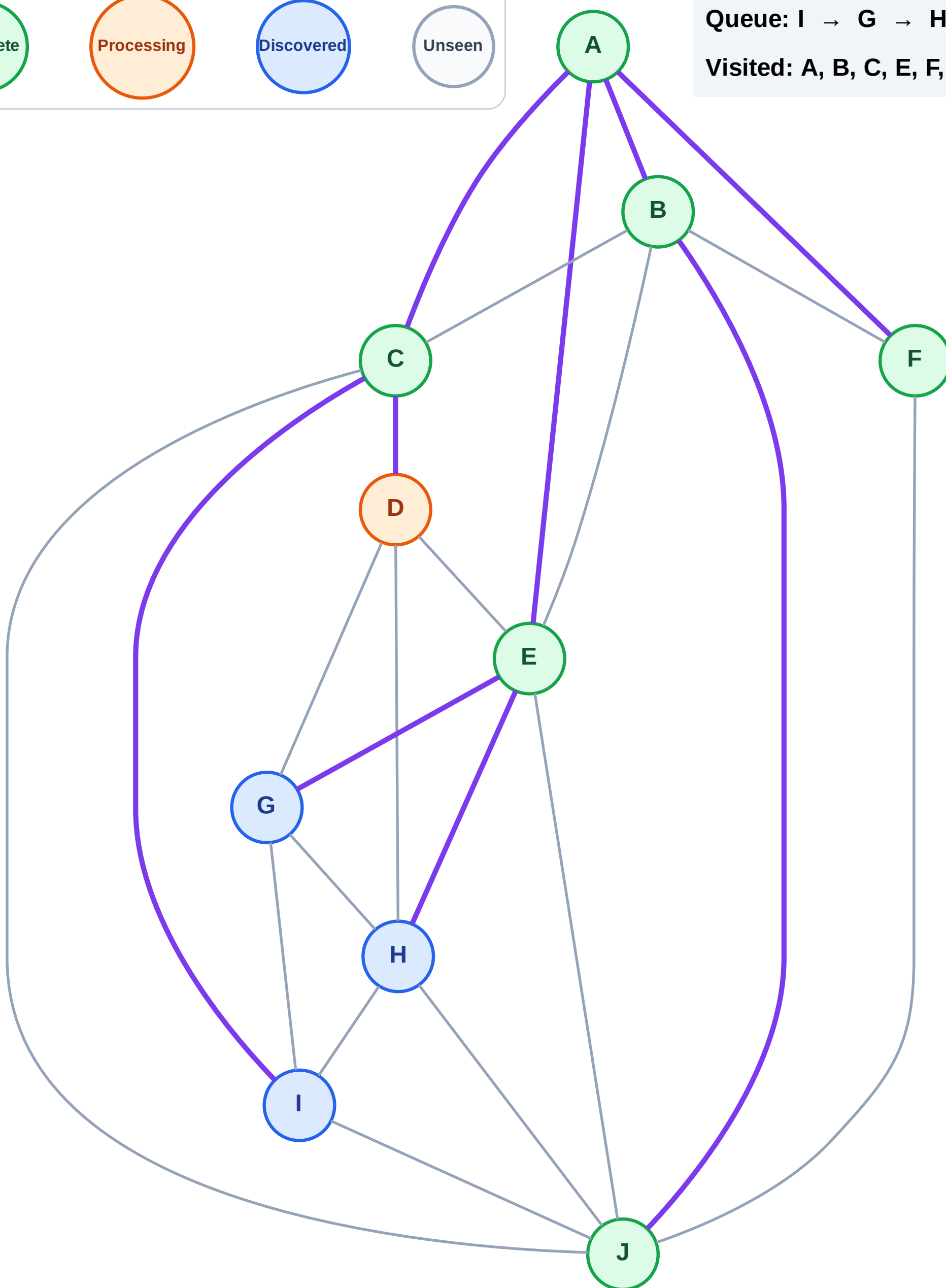


Step 14: Process node D

Legend



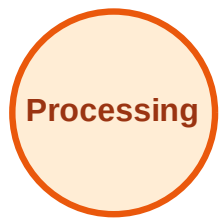
Visited: A, B, C, E, F, J



BFS Traversal

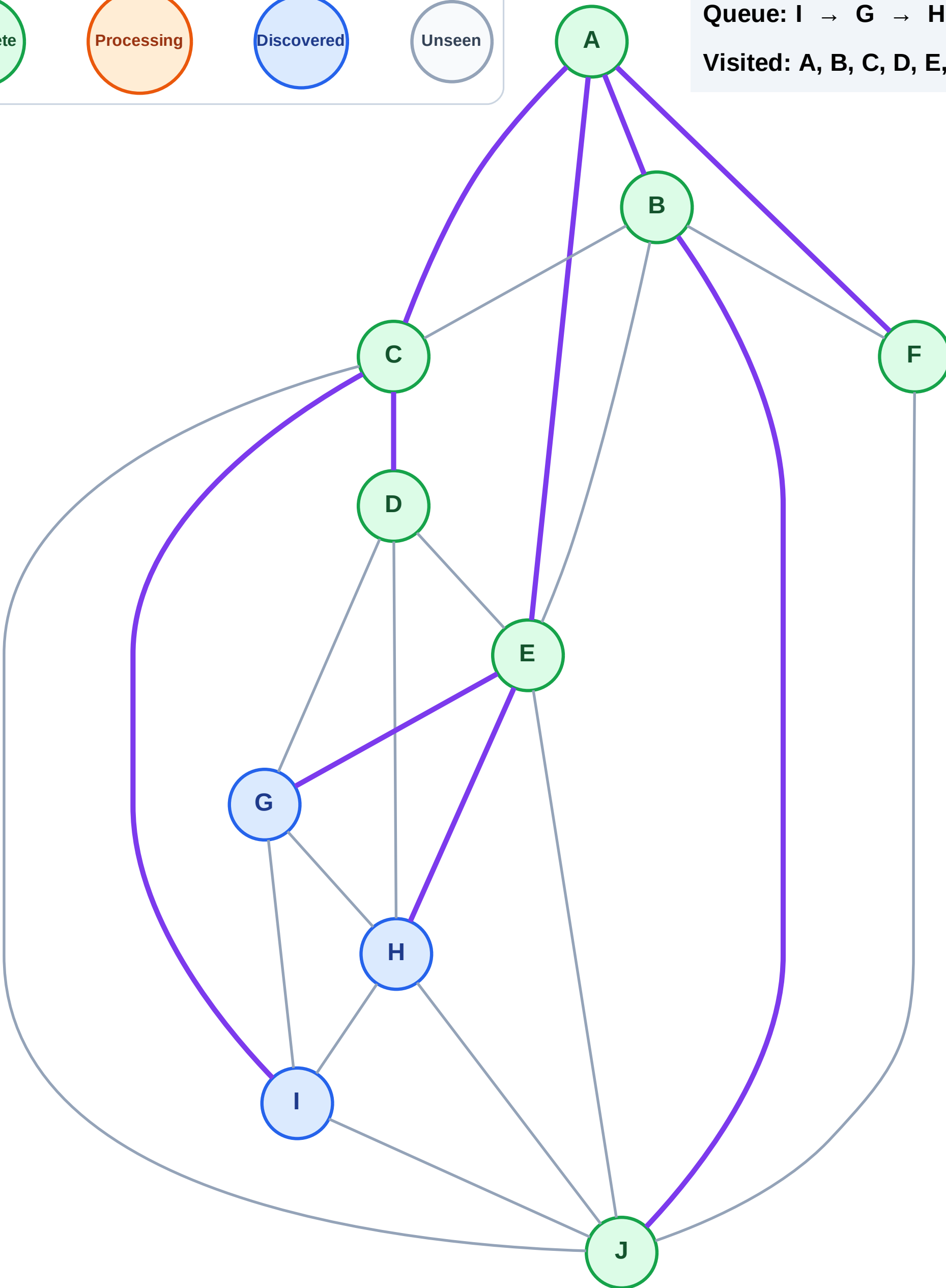
Step 15: No new neighbors from D

Legend



Queue: I → G → H

Visited: A, B, C, D, E, F, J



BFS Traversal

Step 16: Process node I

Legend

Complete

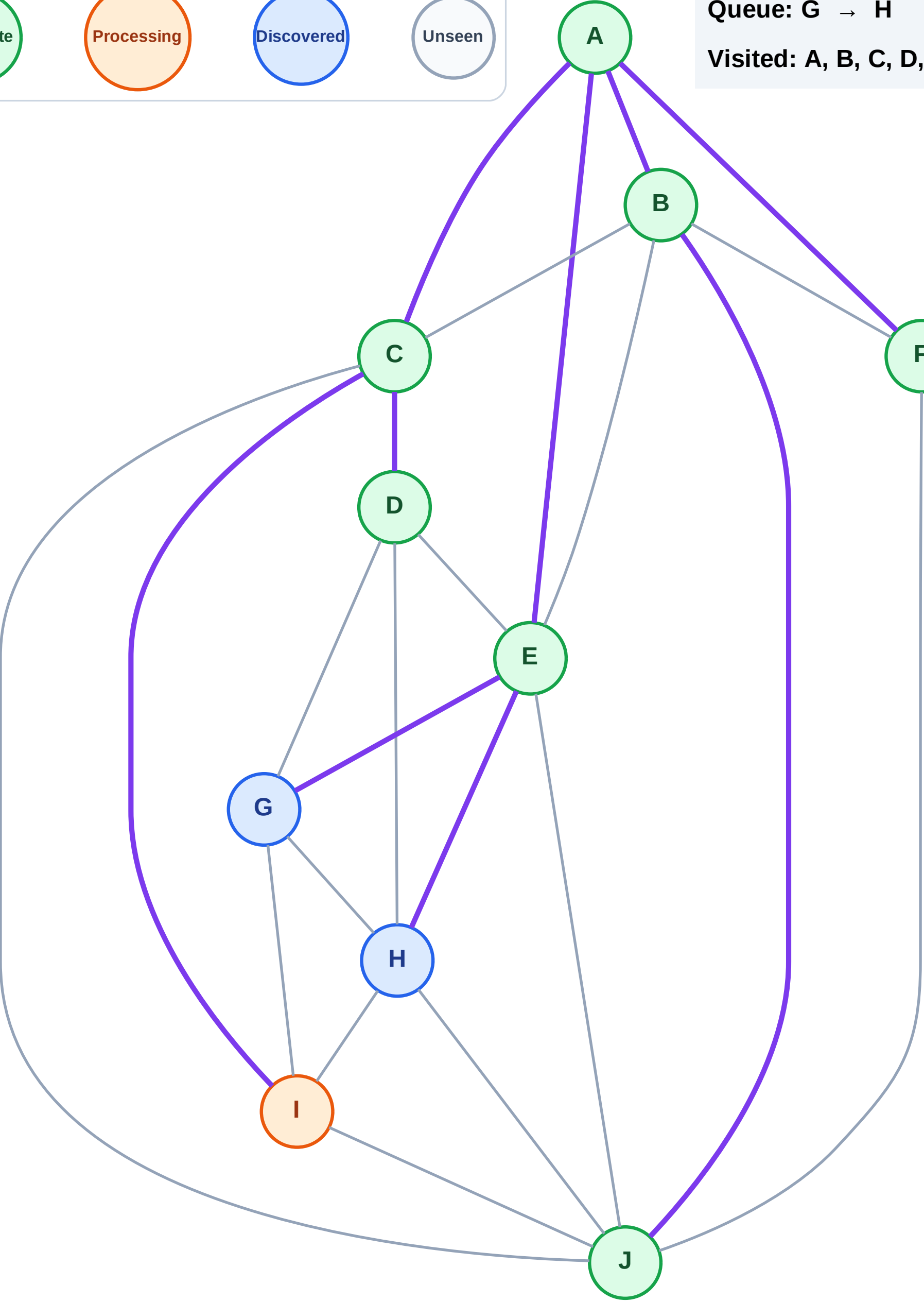
Processing

Discovered

Unseen

Queue: G → H

Visited: A, B, C, D, E, F, J



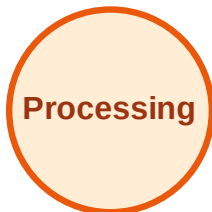
BFS Traversal

Step 17: No new neighbors from I

Legend



Complete



Processing



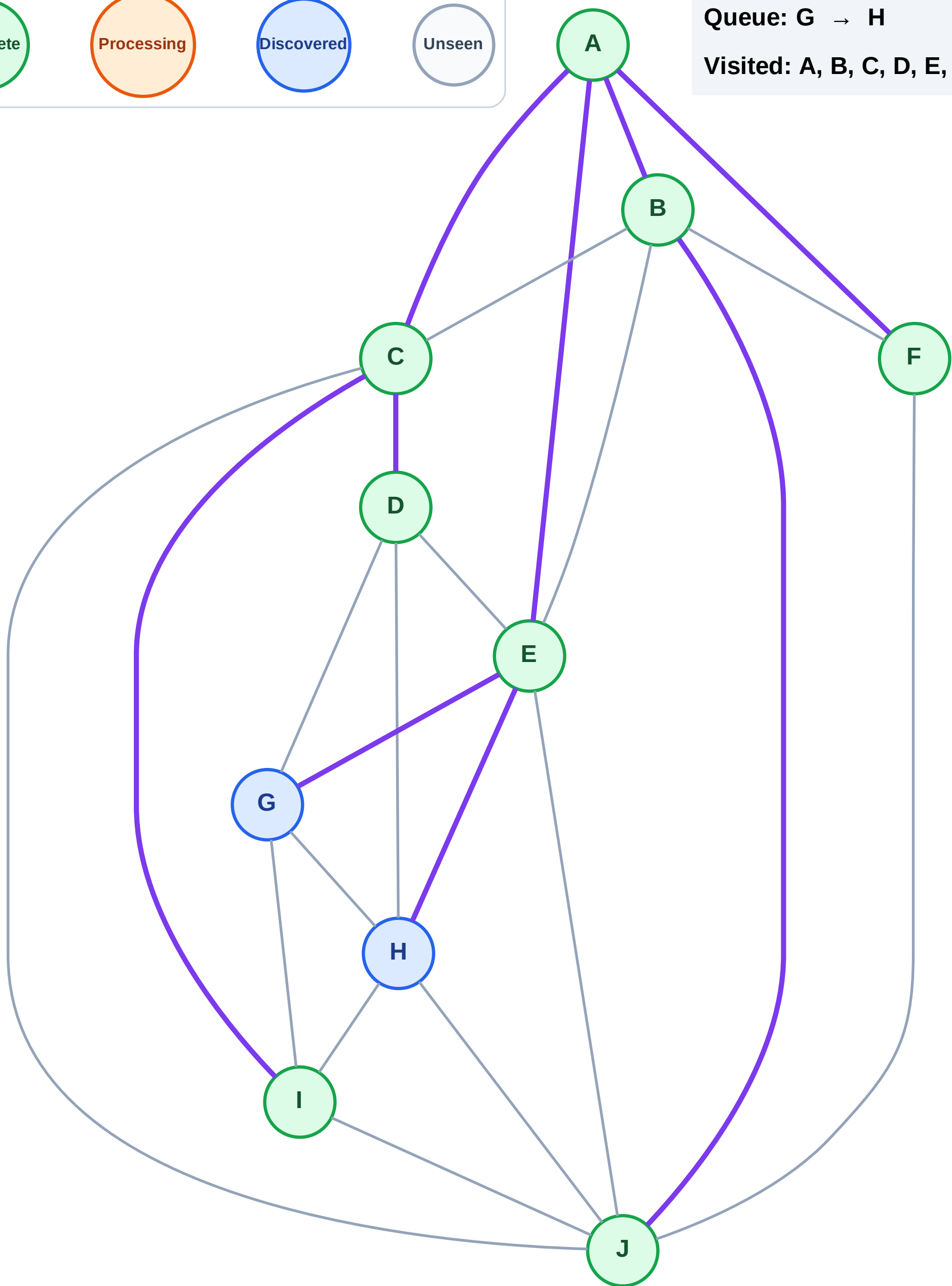
Discovered



Unseen

Queue: G → H

Visited: A, B, C, D, E, F, I, J



BFS Traversal

Step 18: Process node G

Legend

Complete

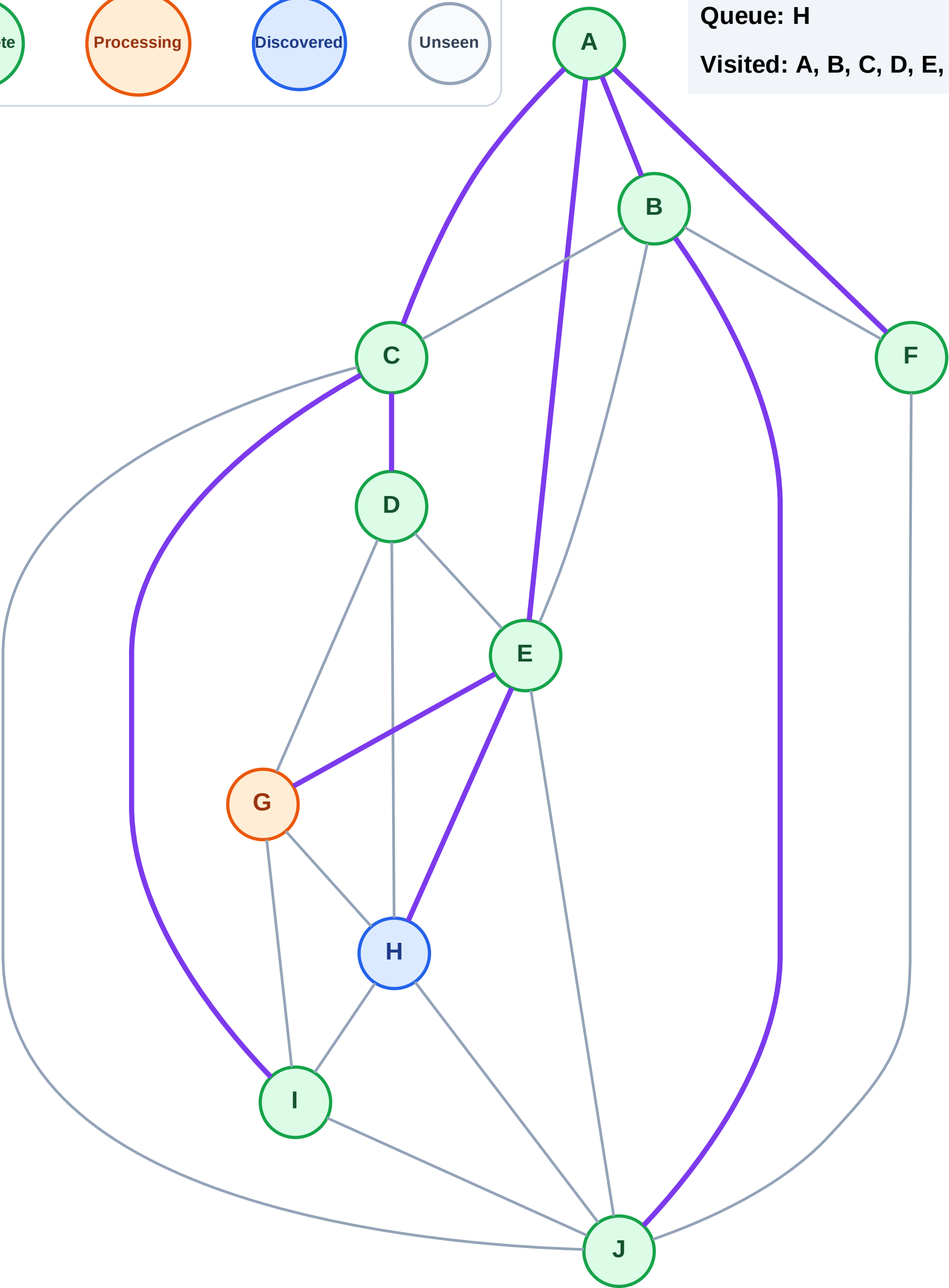
Processing

Discovered

Unseen

Queue: H

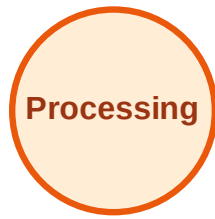
Visited: A, B, C, D, E, F, I, J



BFS Traversal

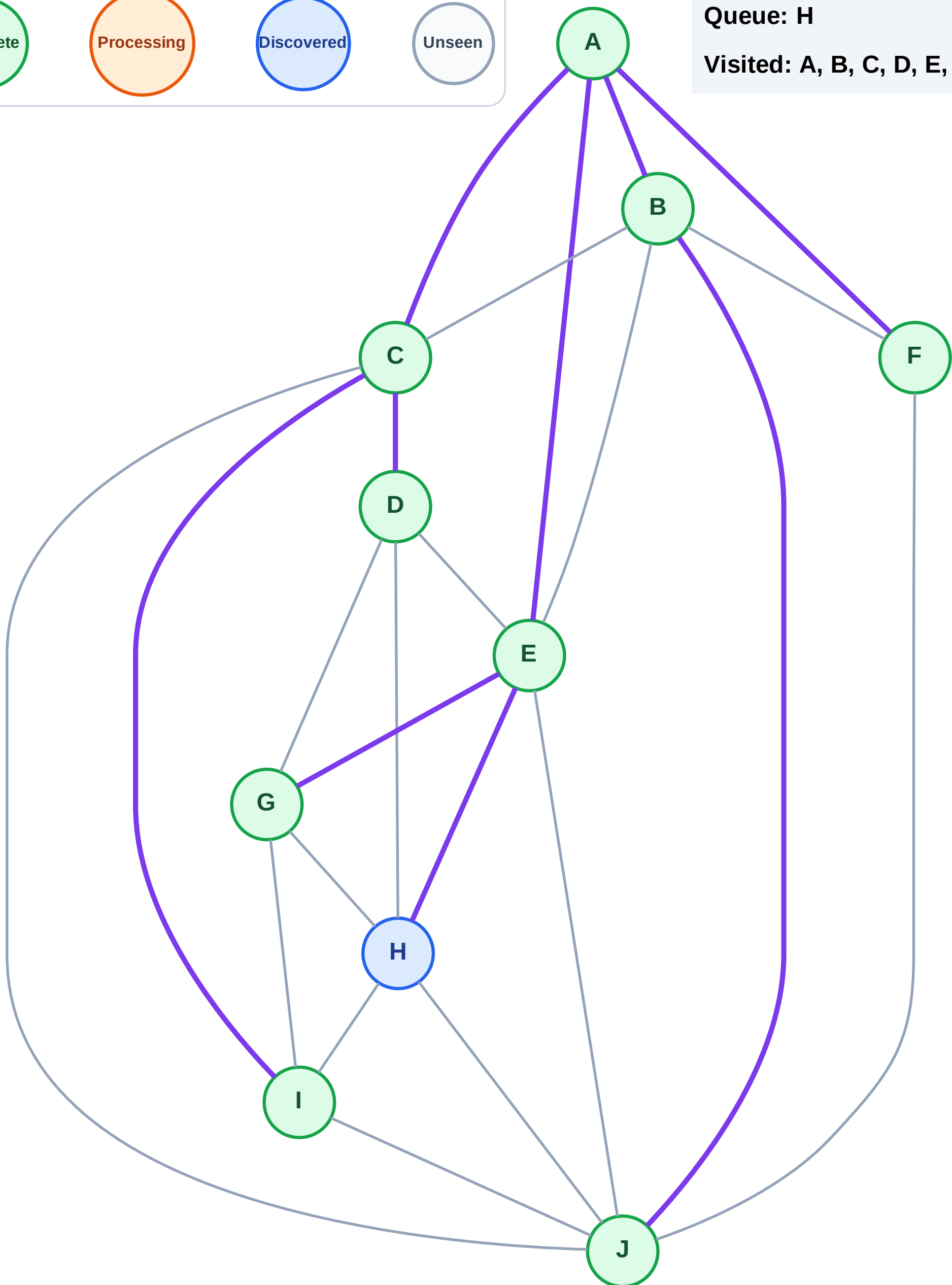
Step 19: No new neighbors from G

Legend



Queue: H

Visited: A, B, C, D, E, F, G, I, J



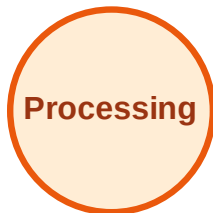
BFS Traversal

Step 20: Process node H

Legend



Complete



Processing



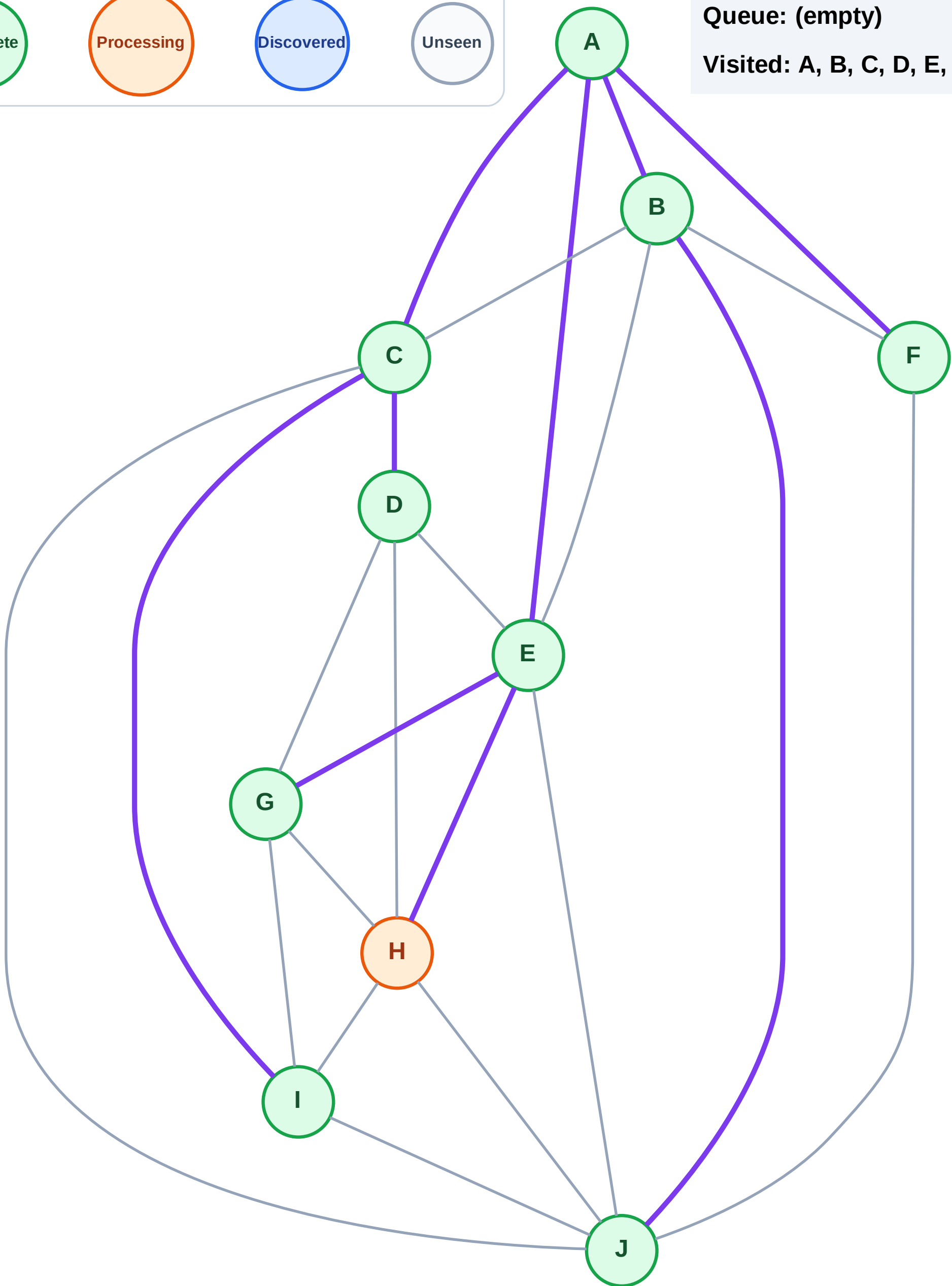
Discovered



Unseen

Queue: (empty)

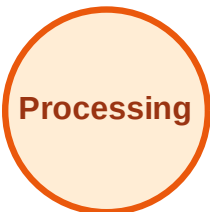
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

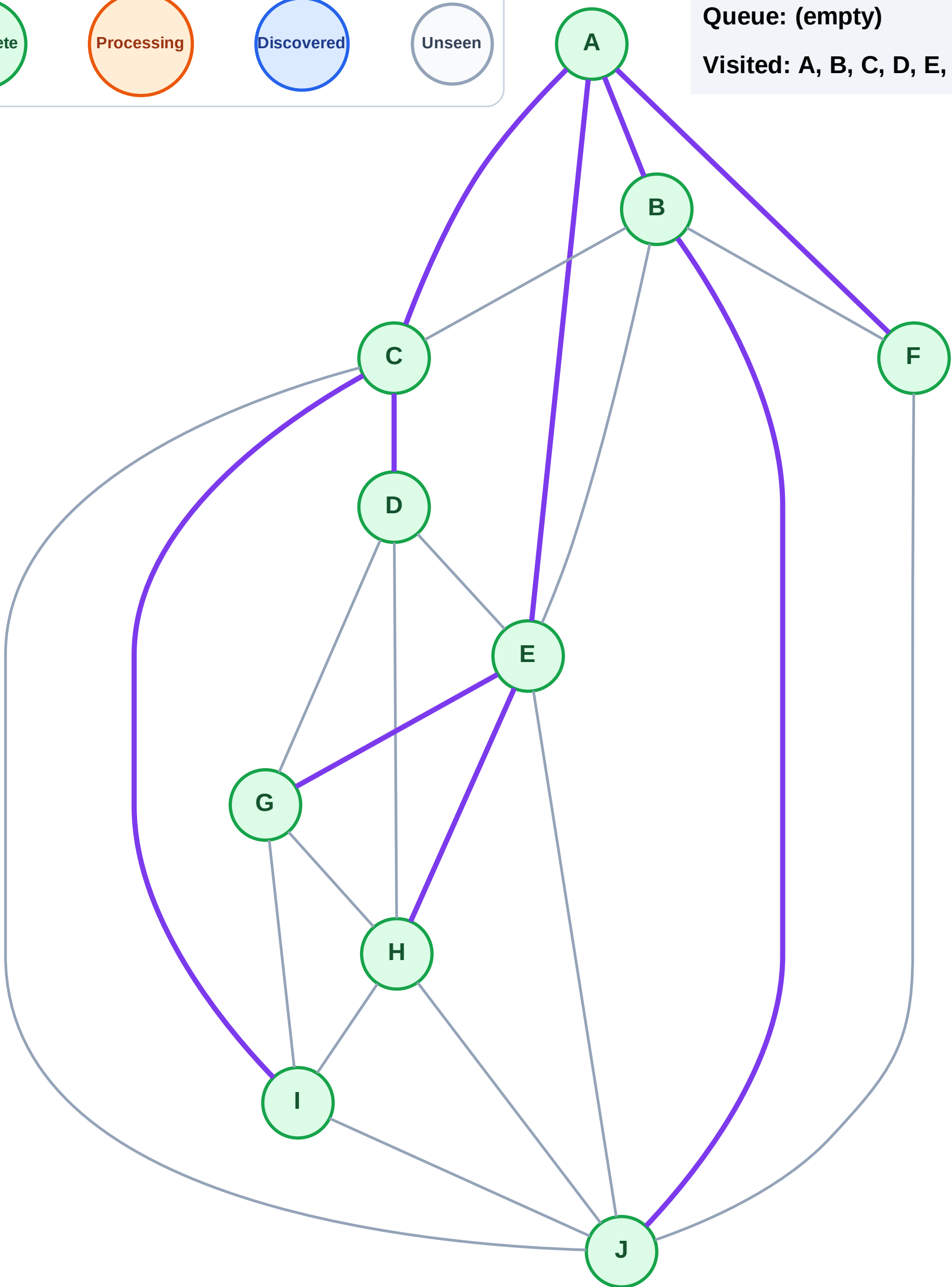
Step 21: No new neighbors from H

Legend



Queue: (empty)

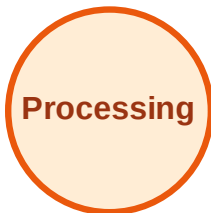
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

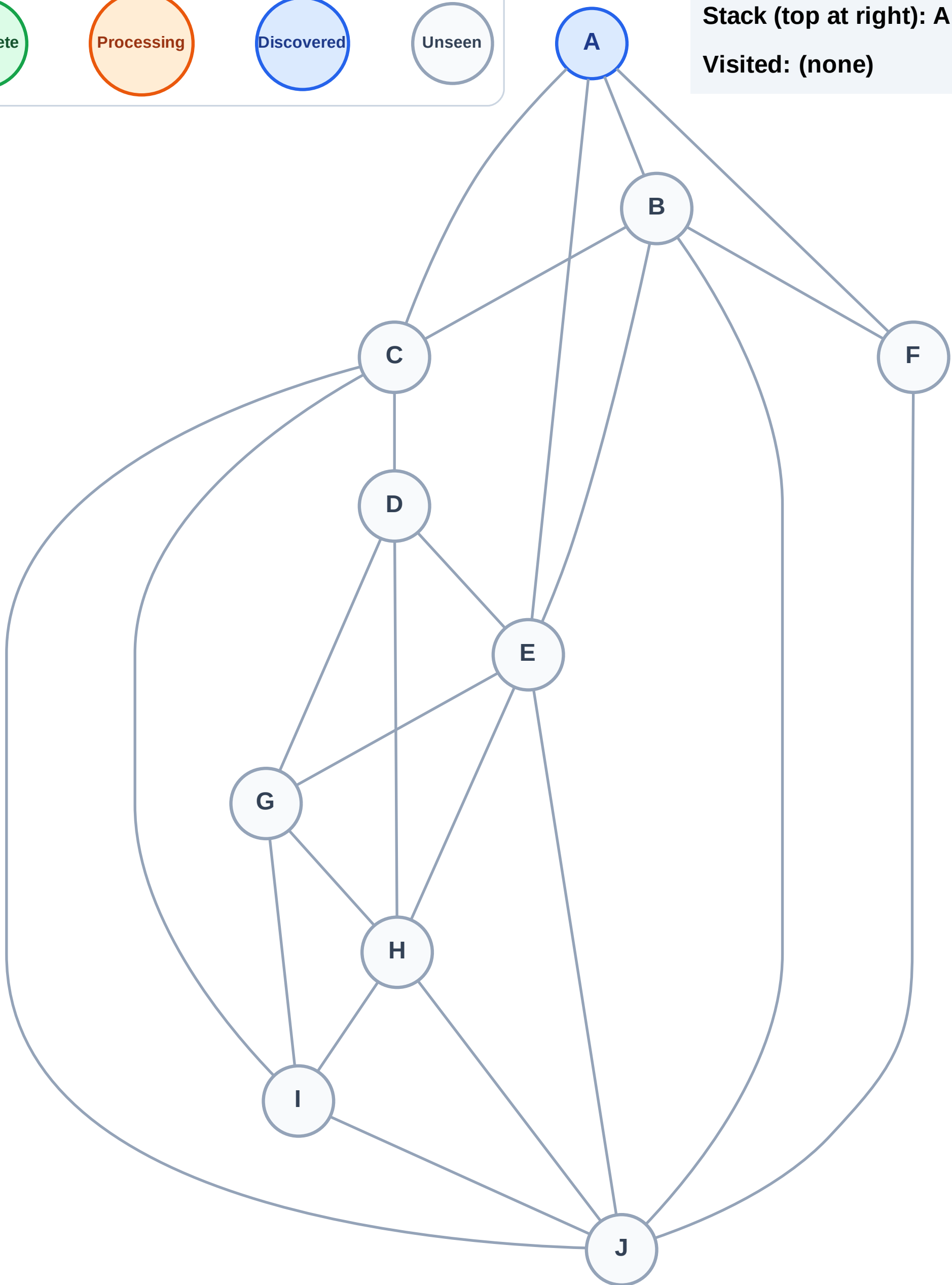
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

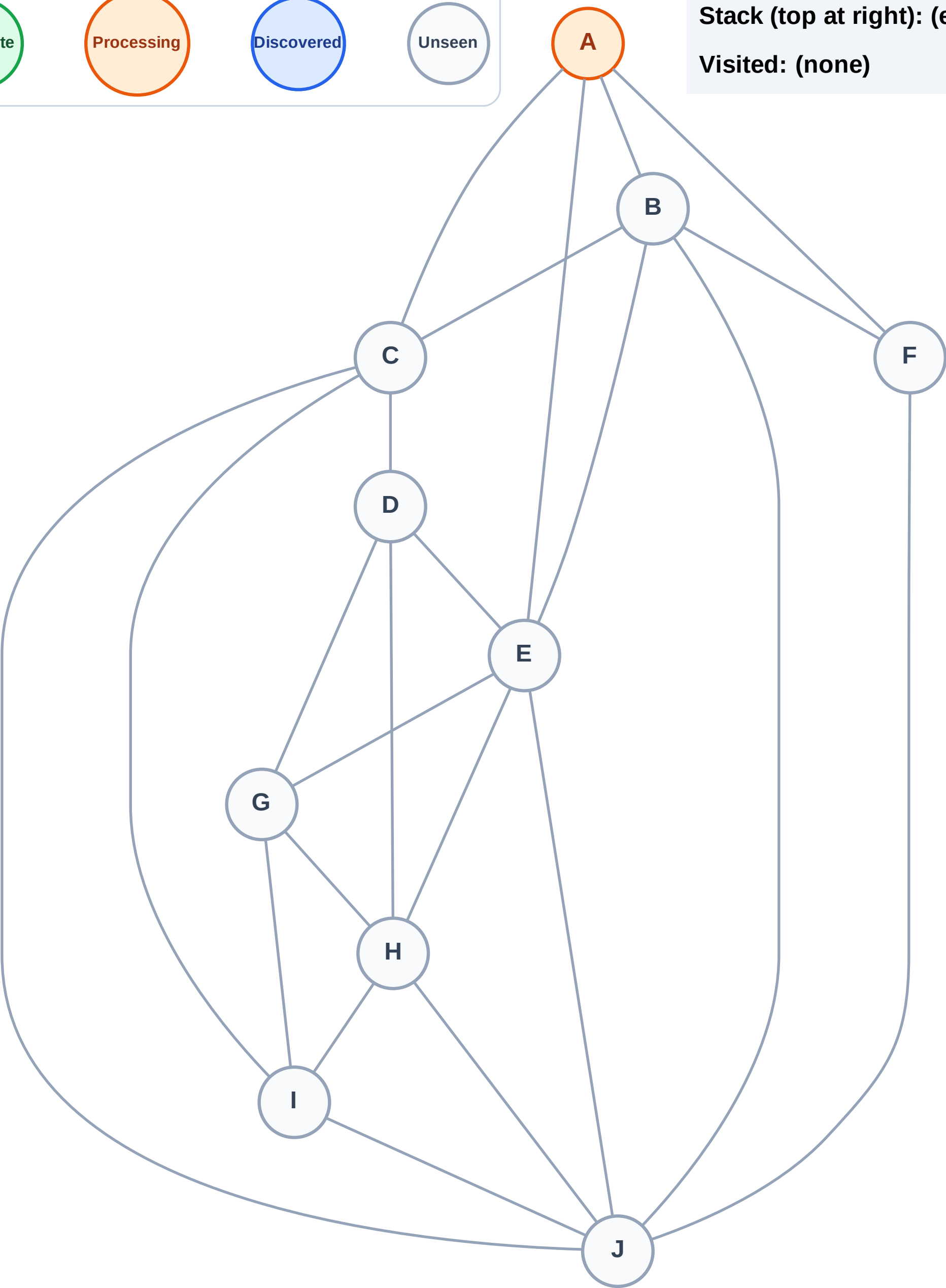
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover F, E, C, B from A

Legend

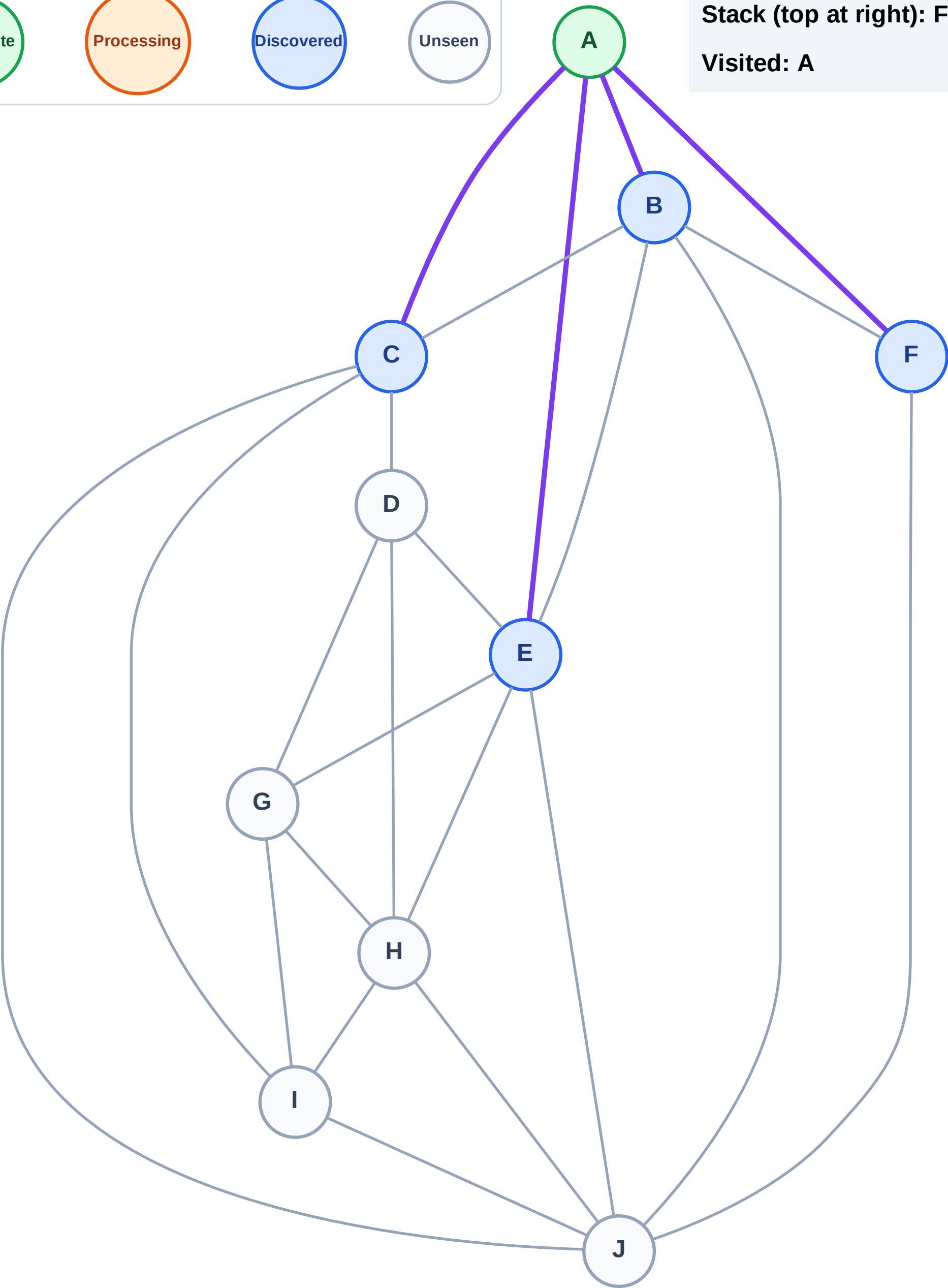
Complete

Processing

Discovered

Unseen

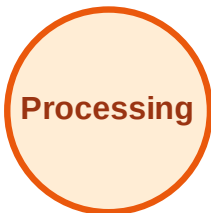
Stack (top at right): F | E | C | B
Visited: A



DFS Traversal

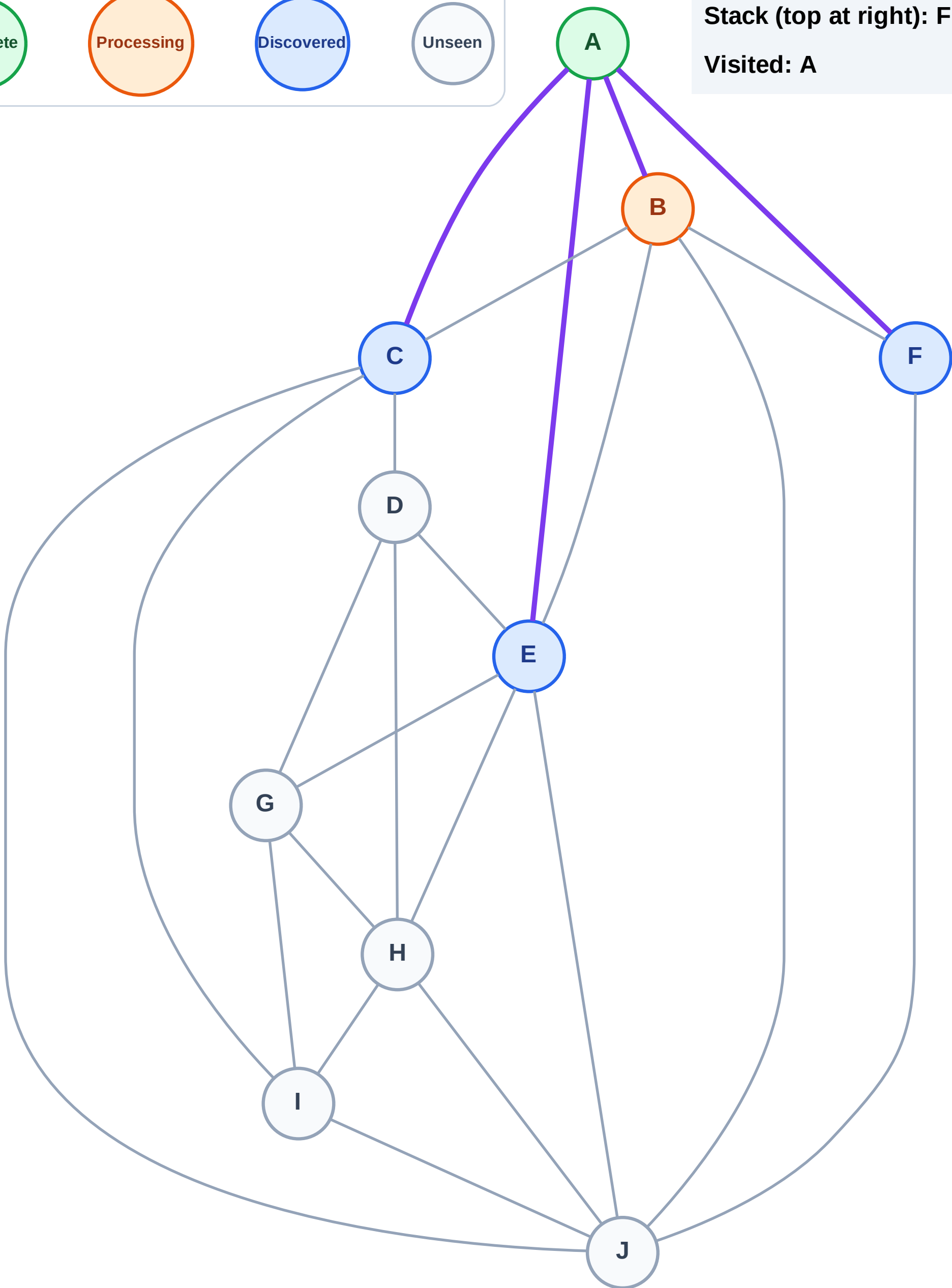
Step 4: Process node B

Legend



Stack (top at right): F | E | C

Visited: A



DFS Traversal

Step 5: Discover J from B

Legend

Complete

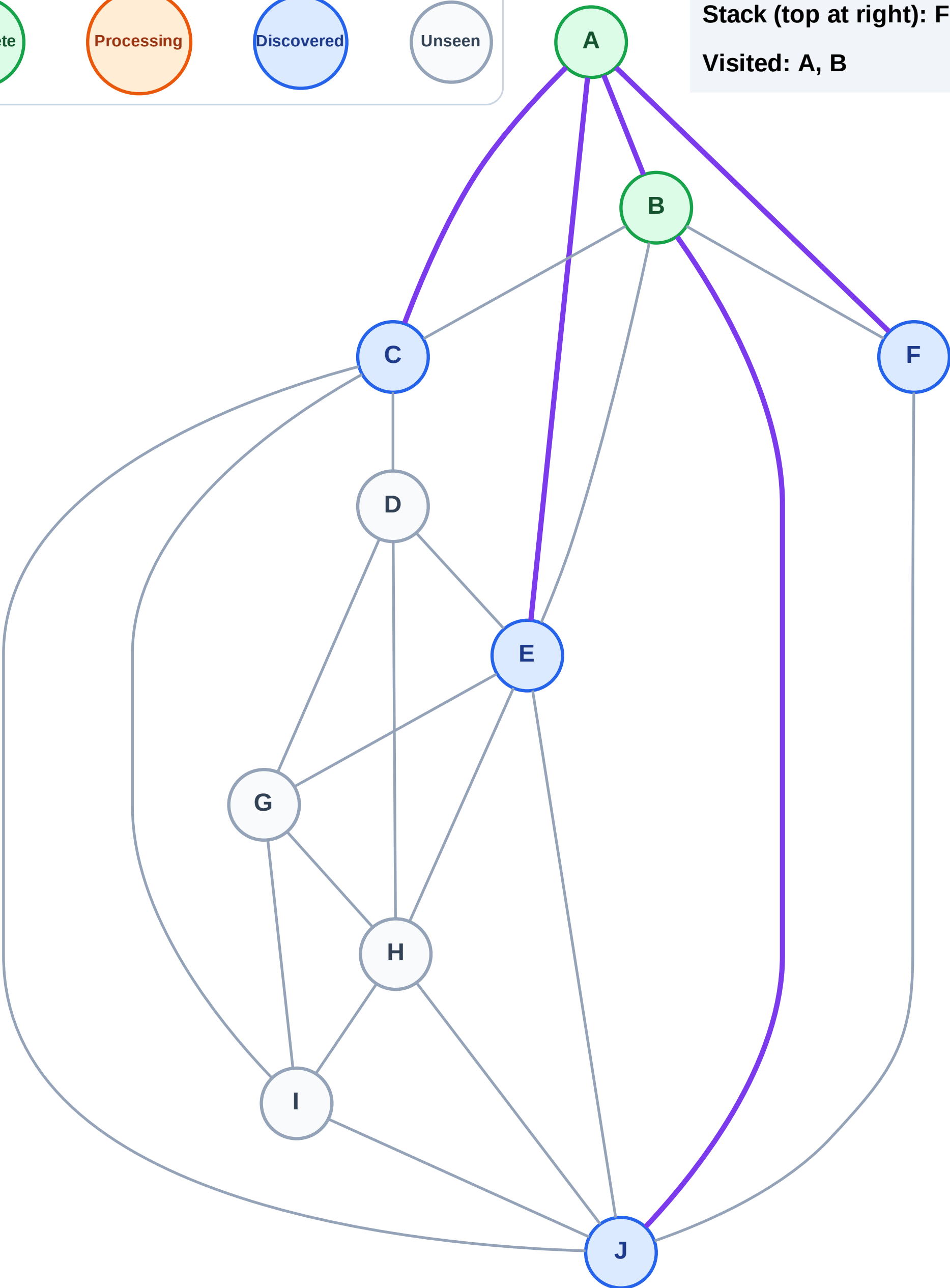
Processing

Discovered

Unseen

Stack (top at right): F | E | C | J

Visited: A, B



DFS Traversal

Step 6: Process node J

Legend

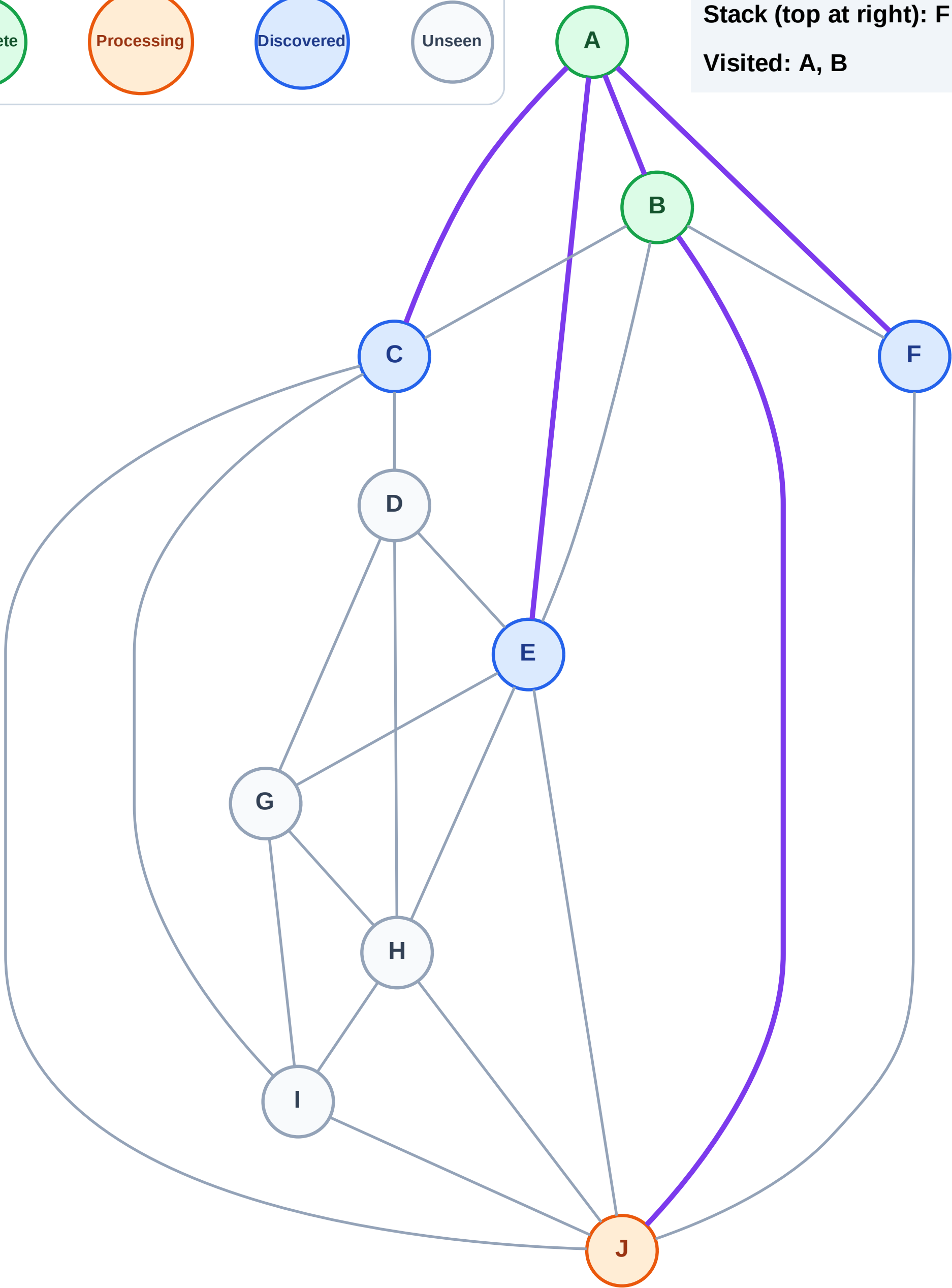
Complete

Processing

Discovered

Unseen

Stack (top at right): F | E | C
Visited: A, B



DFS Traversal

Step 7: Discover I, H from J

Legend

Complete

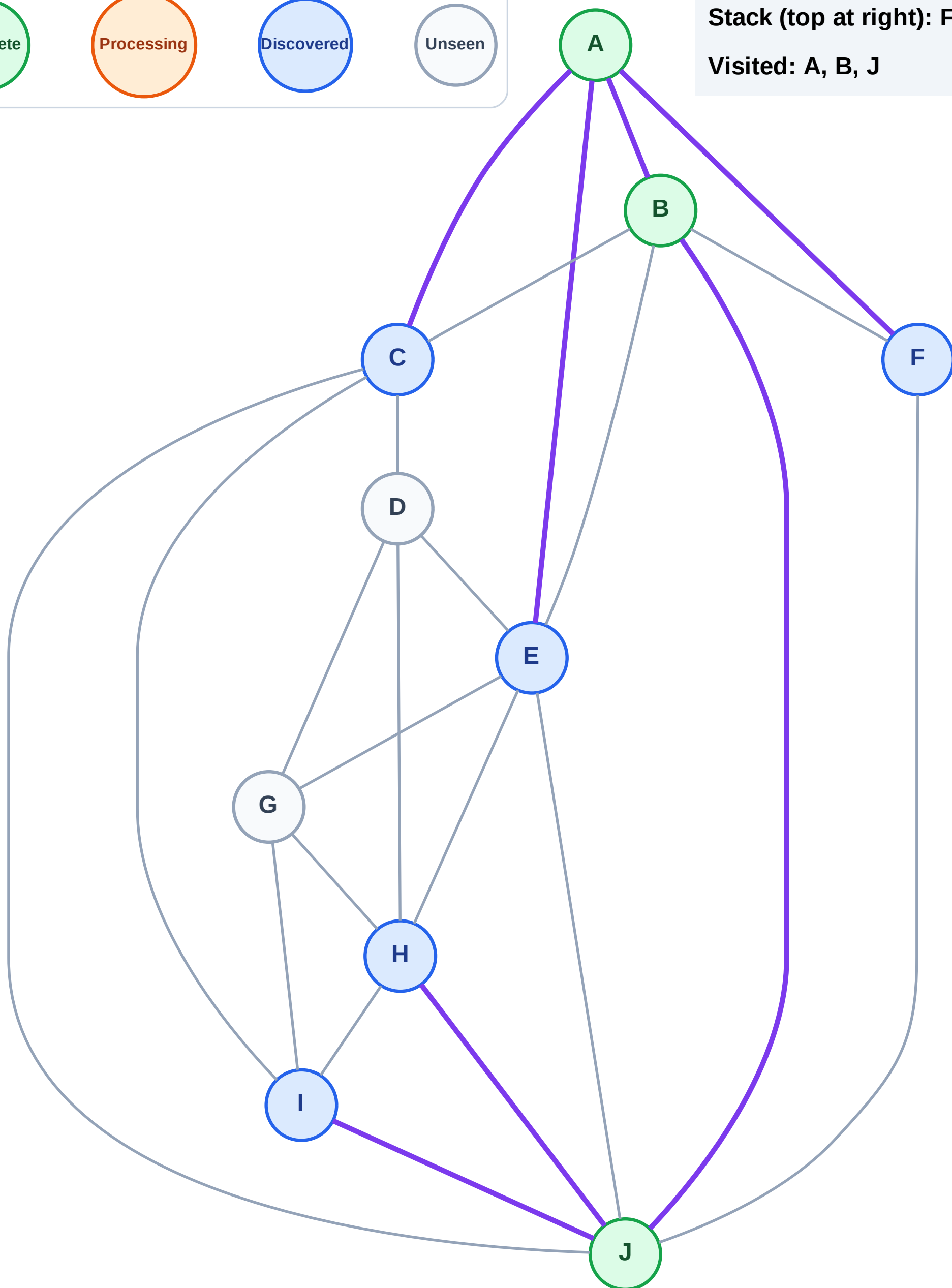
Processing

Discovered

Unseen

Stack (top at right): F | E | C | I | H

Visited: A, B, J



DFS Traversal

Step 8: Process node H

Legend

Complete

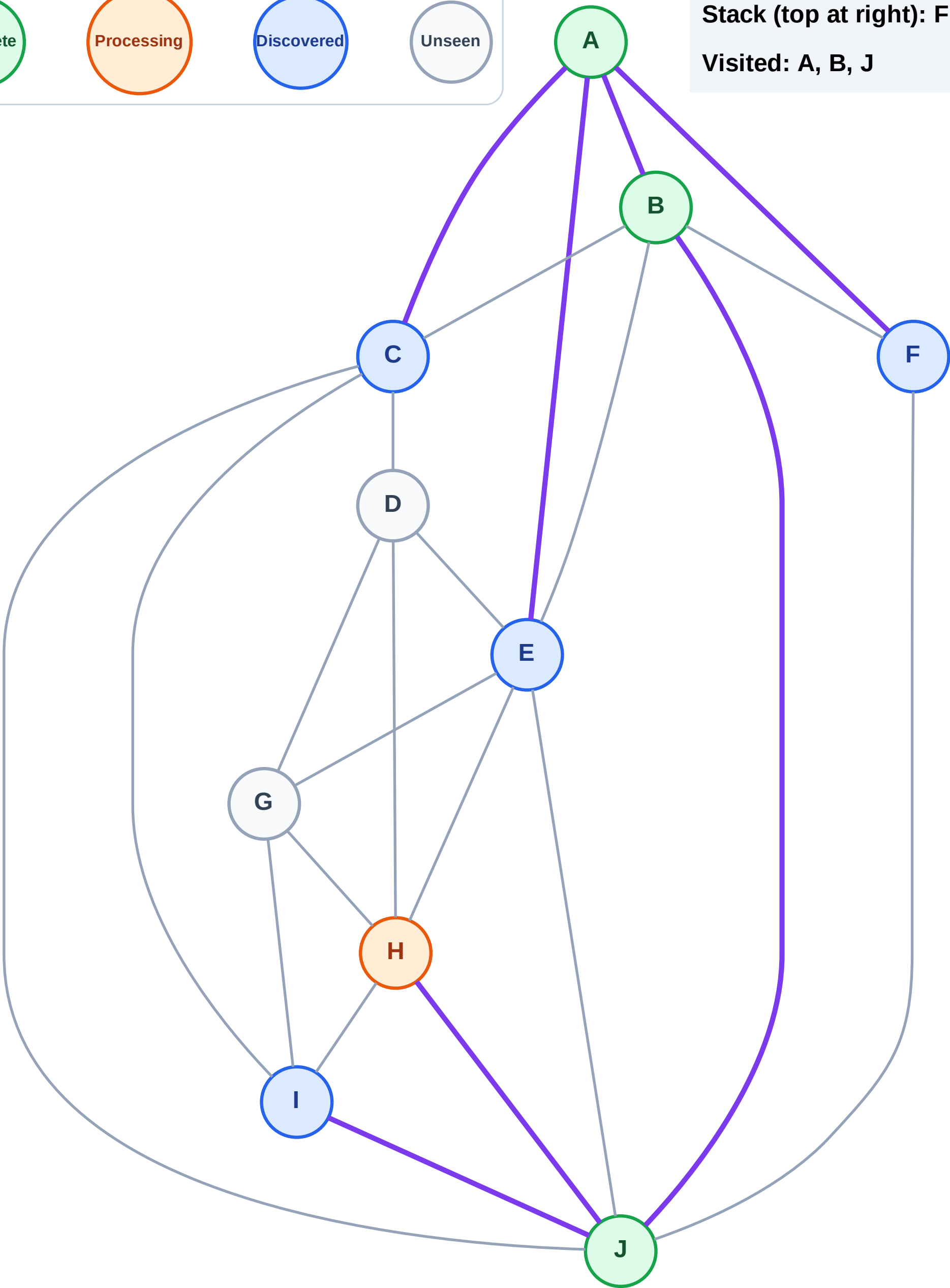
Processing

Discovered

Unseen

Stack (top at right): F | E | C | I

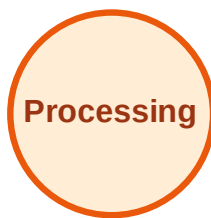
Visited: A, B, J



DFS Traversal

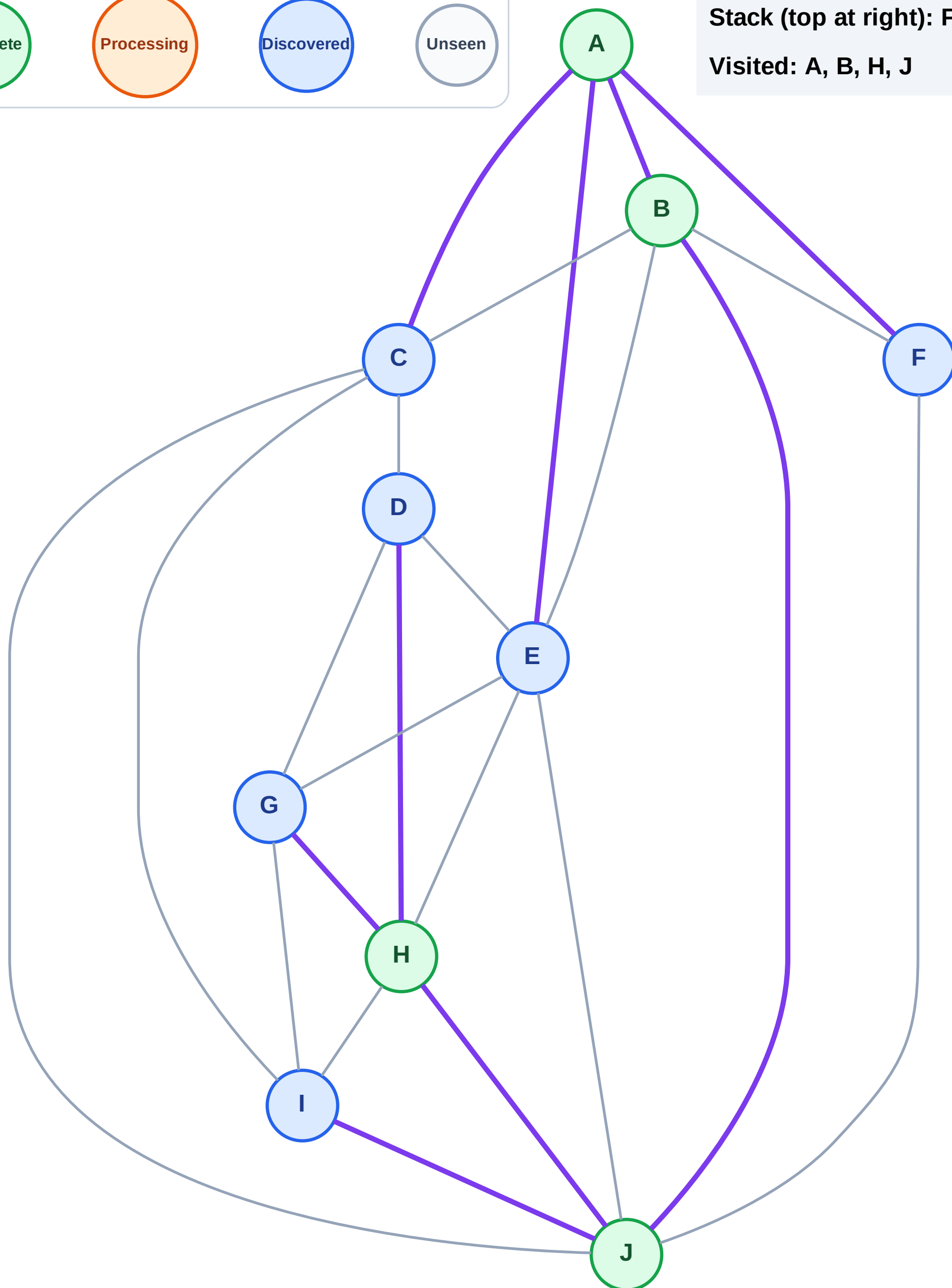
Step 9: Discover G, D from H

Legend



Stack (top at right): F | E | C | I | G | D

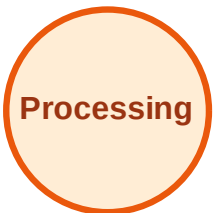
Visited: A, B, H, J



DFS Traversal

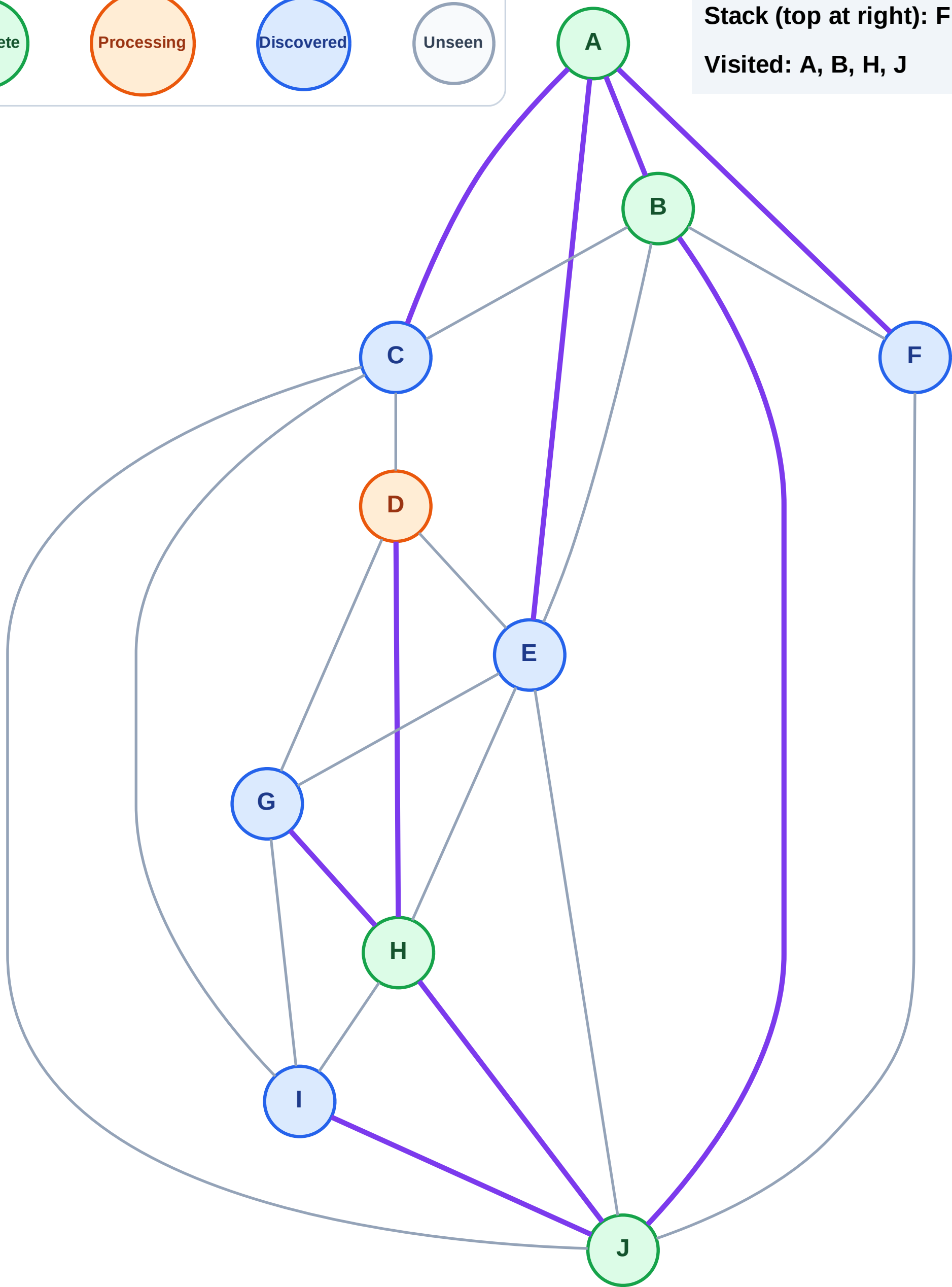
Step 10: Process node D

Legend



Stack (top at right): F | E | C | I | G

Visited: A, B, H, J



DFS Traversal

Step 11: No new neighbors from D

Legend

Complete

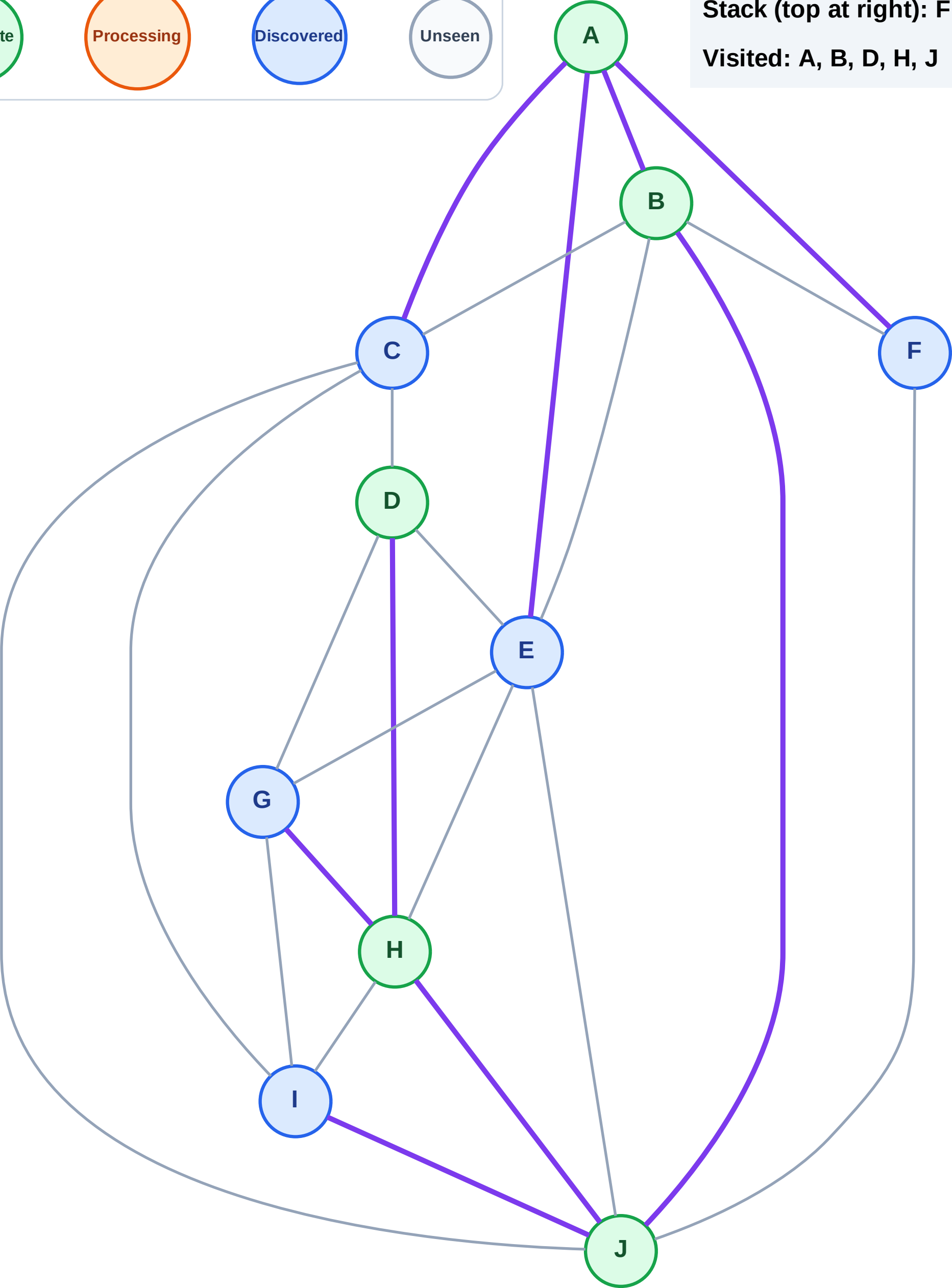
Processing

Discovered

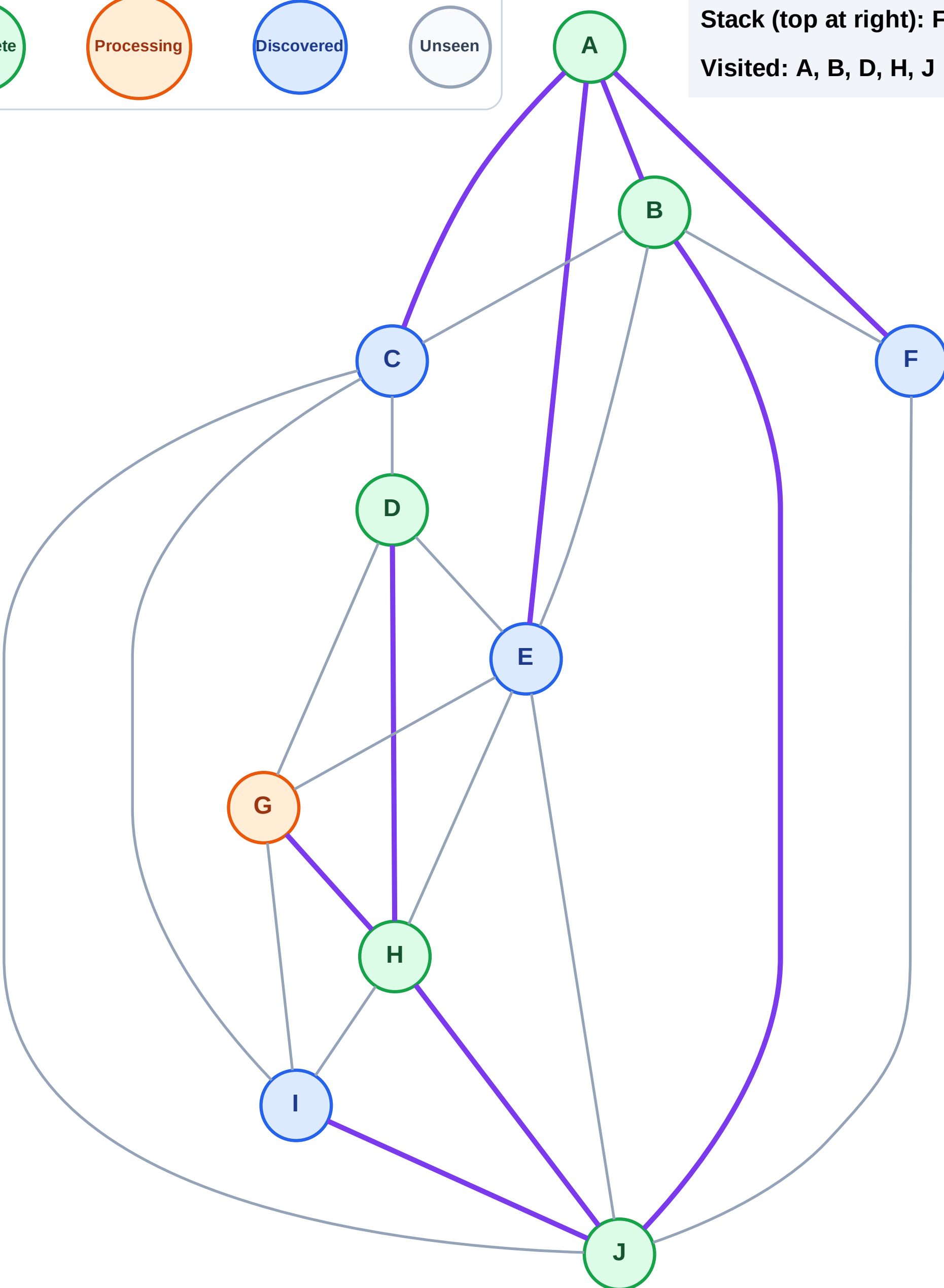
Unseen

Stack (top at right): F | E | C | I | G

Visited: A, B, D, H, J



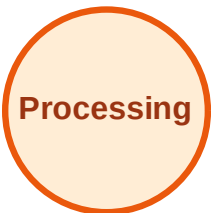
Step 12: Process node G



DFS Traversal

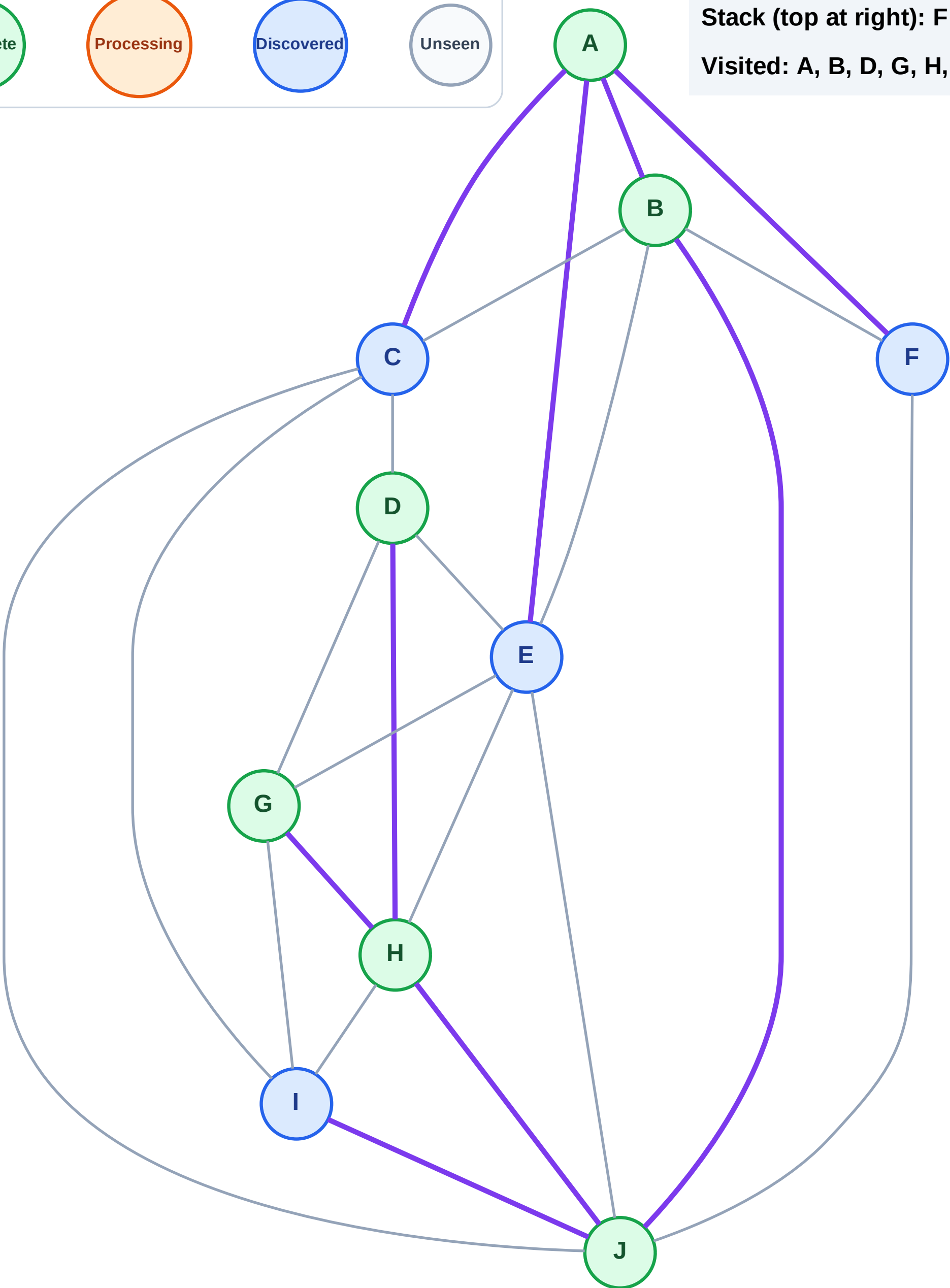
Step 13: No new neighbors from G

Legend



Stack (top at right): F | E | C | I

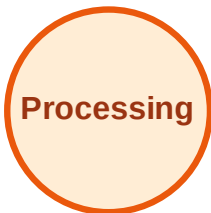
Visited: A, B, D, G, H, J



DFS Traversal

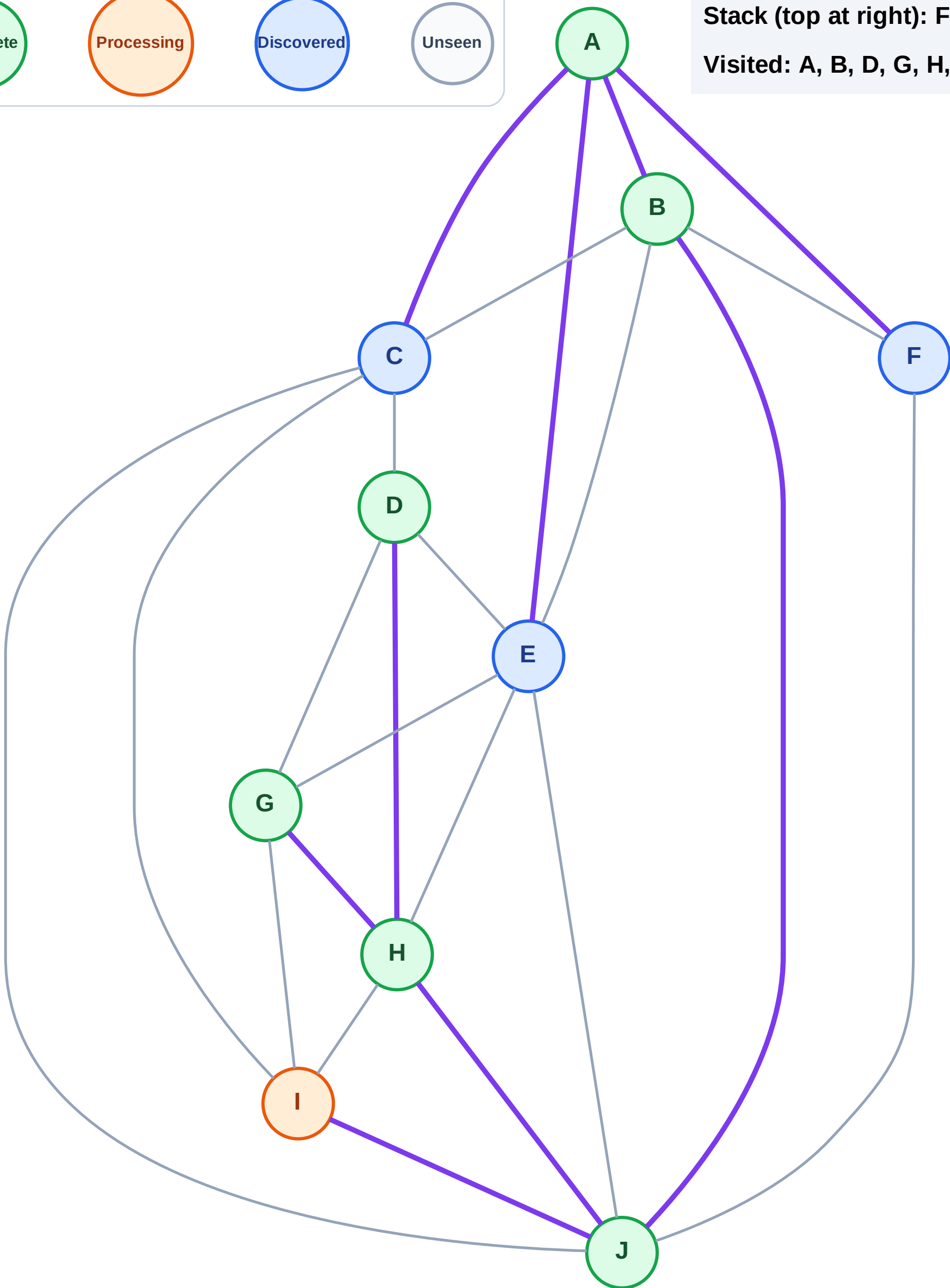
Step 14: Process node I

Legend



Stack (top at right): F | E | C

Visited: A, B, D, G, H, J



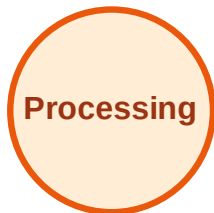
DFS Traversal

Step 15: No new neighbors from I

Legend



Complete



Processing



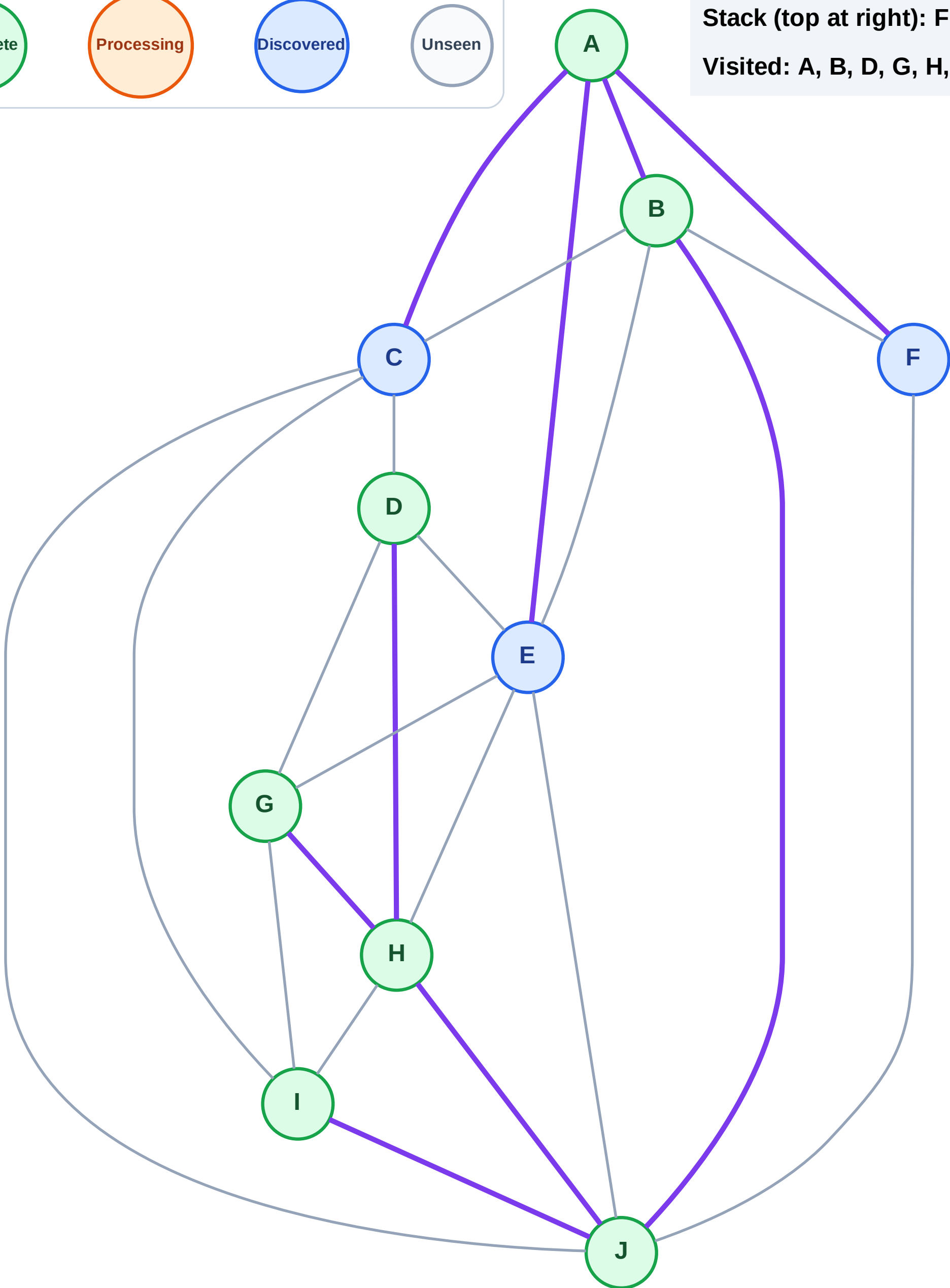
Discovered



Unseen

Stack (top at right): F | E | C

Visited: A, B, D, G, H, I, J



DFS Traversal

Step 16: Process node C

Legend

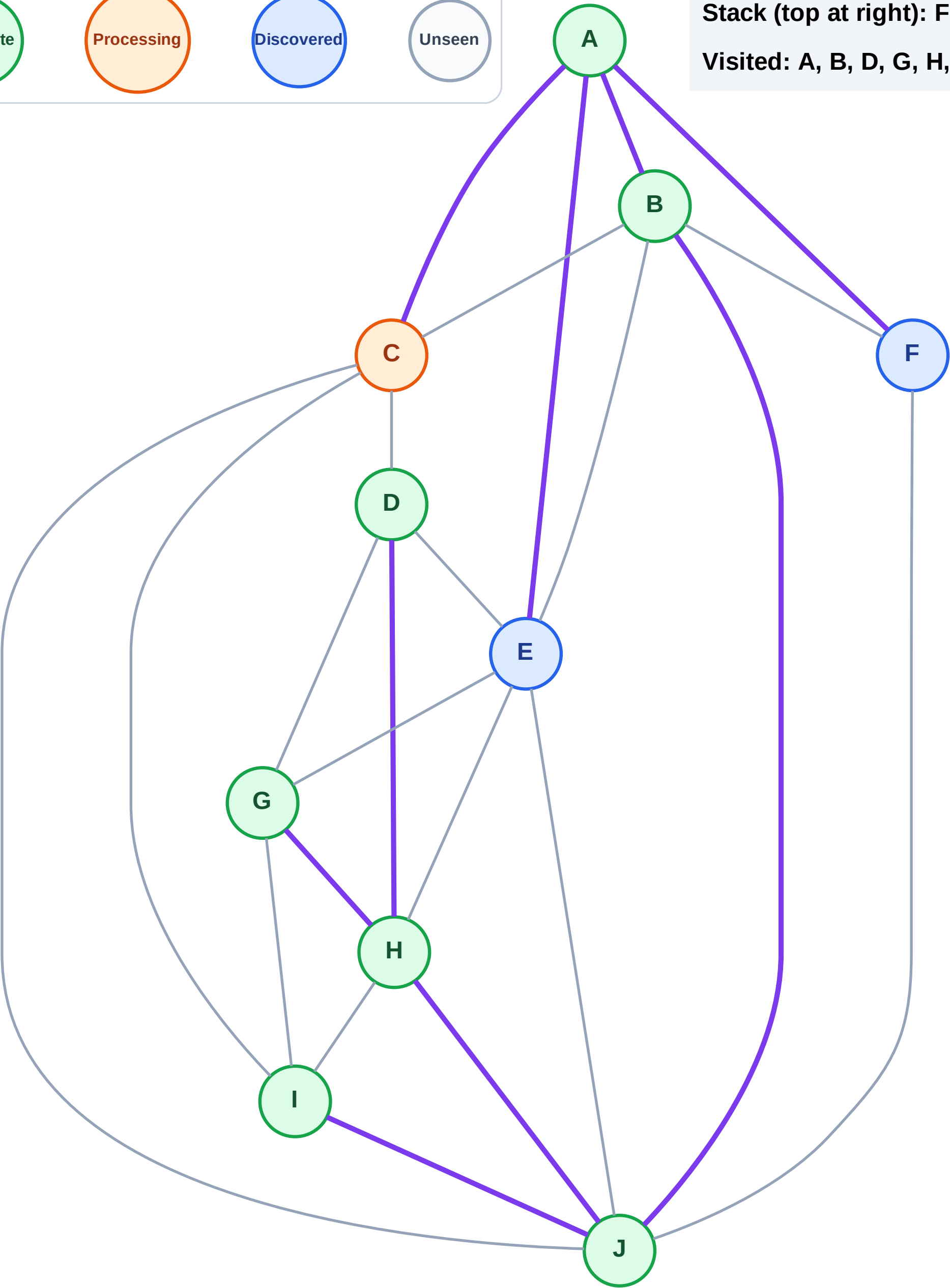
Complete

Processing

Discovered

Unseen

Stack (top at right): F | E
Visited: A, B, D, G, H, I, J



DFS Traversal

Step 17: No new neighbors from C

Legend

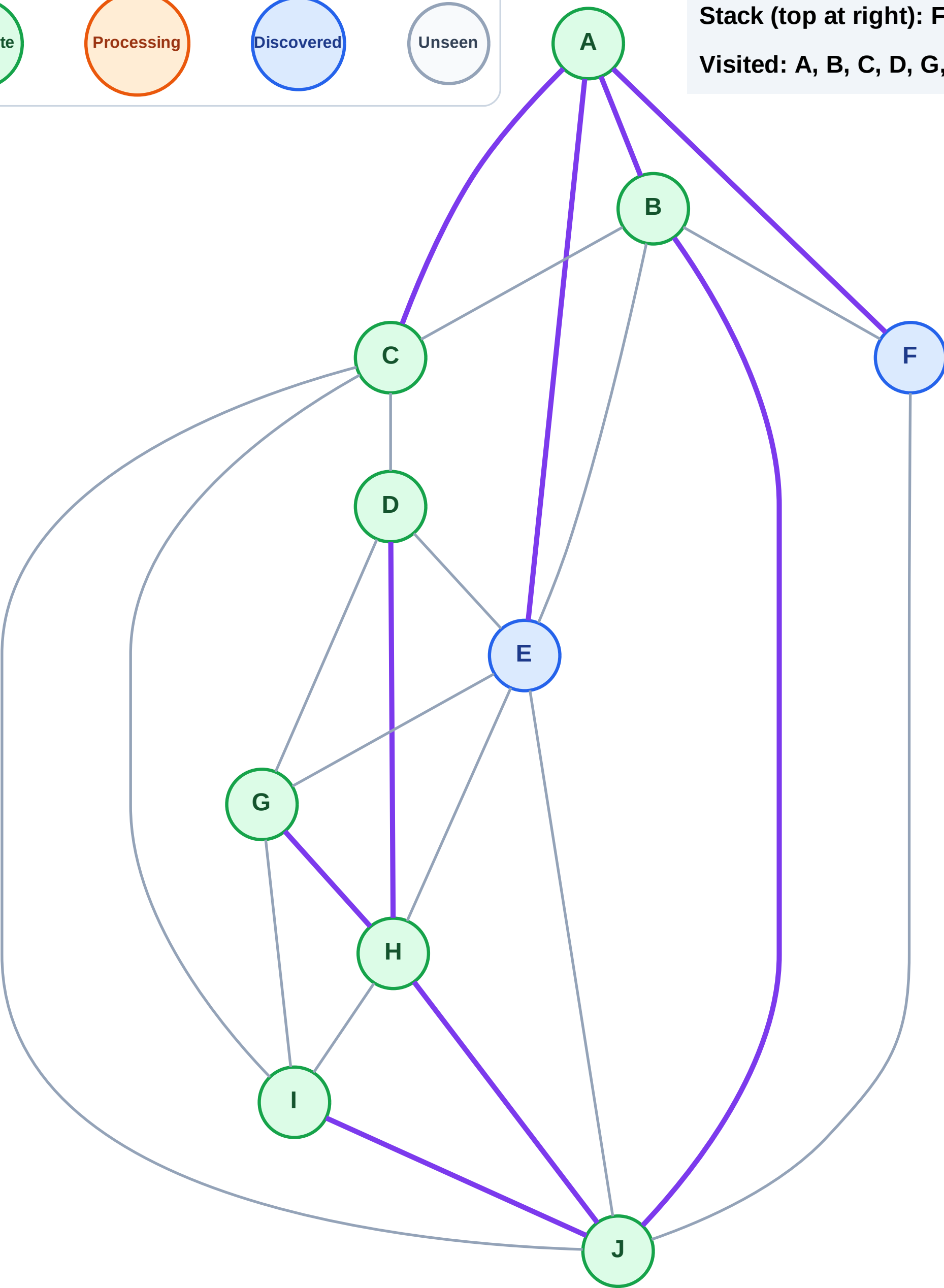
Complete

Processing

Discovered

Unseen

Stack (top at right): F | E
Visited: A, B, C, D, G, H, I, J



DFS Traversal

Step 18: Process node E

Legend

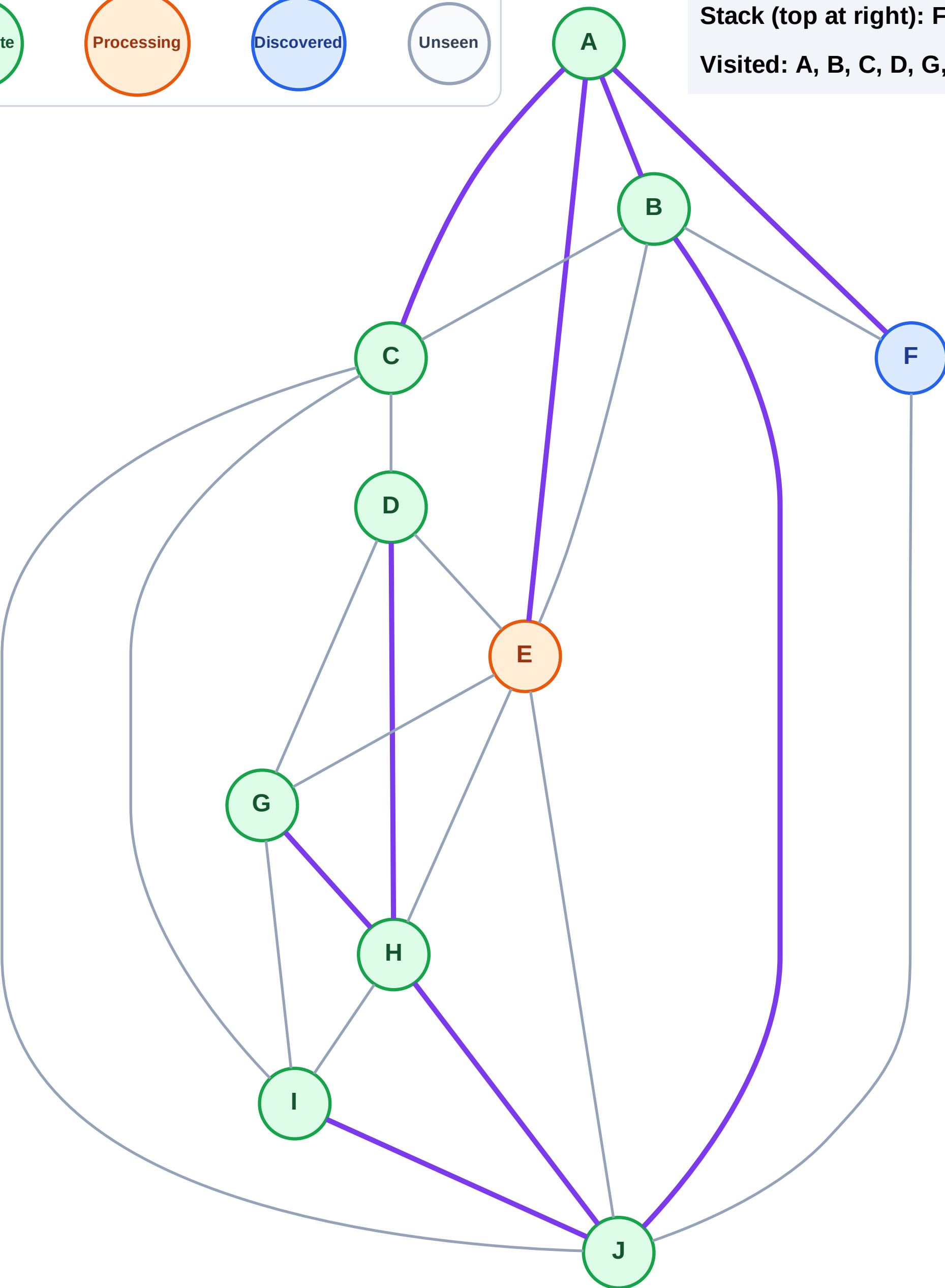
Complete

Processing

Discovered

Unseen

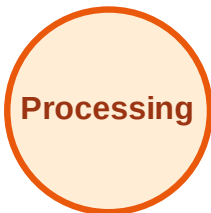
Stack (top at right): F
Visited: A, B, C, D, G, H, I, J



DFS Traversal

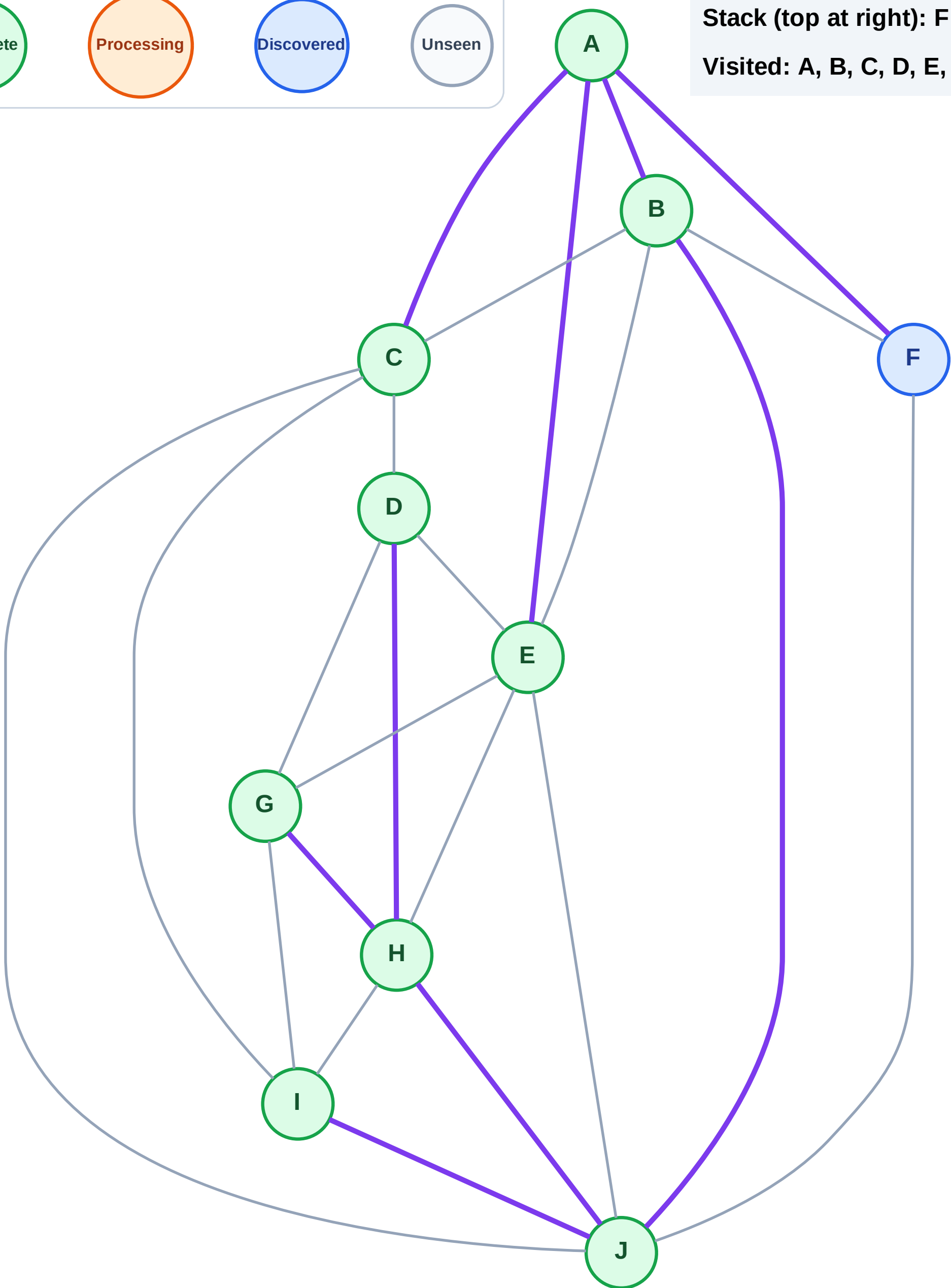
Step 19: No new neighbors from E

Legend



Stack (top at right): F

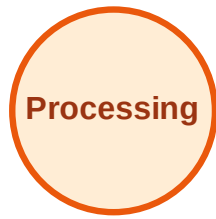
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

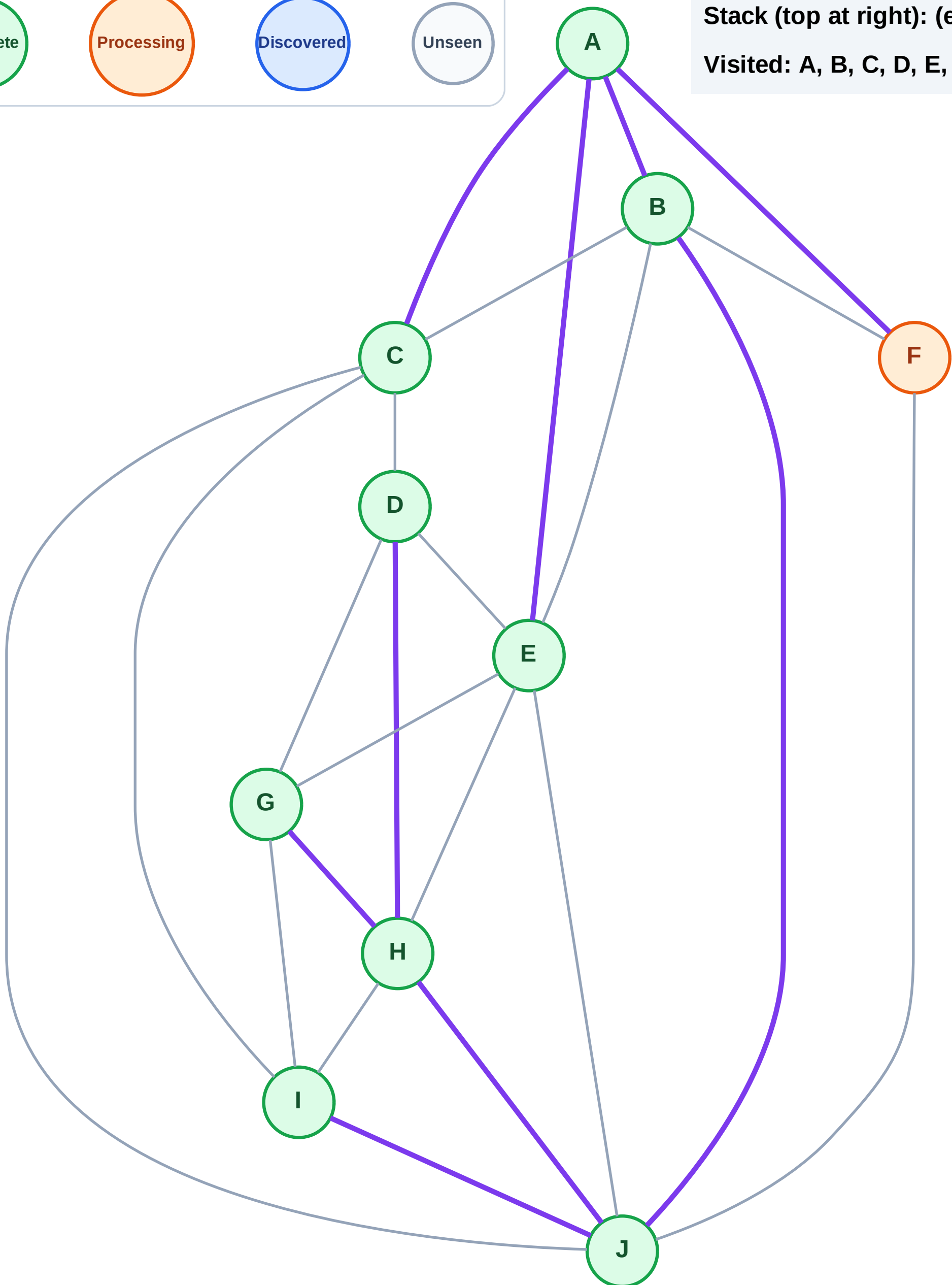
Step 20: Process node F

Legend



Stack (top at right): (empty)

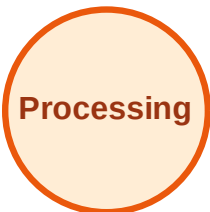
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

Step 21: No new neighbors from F

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

